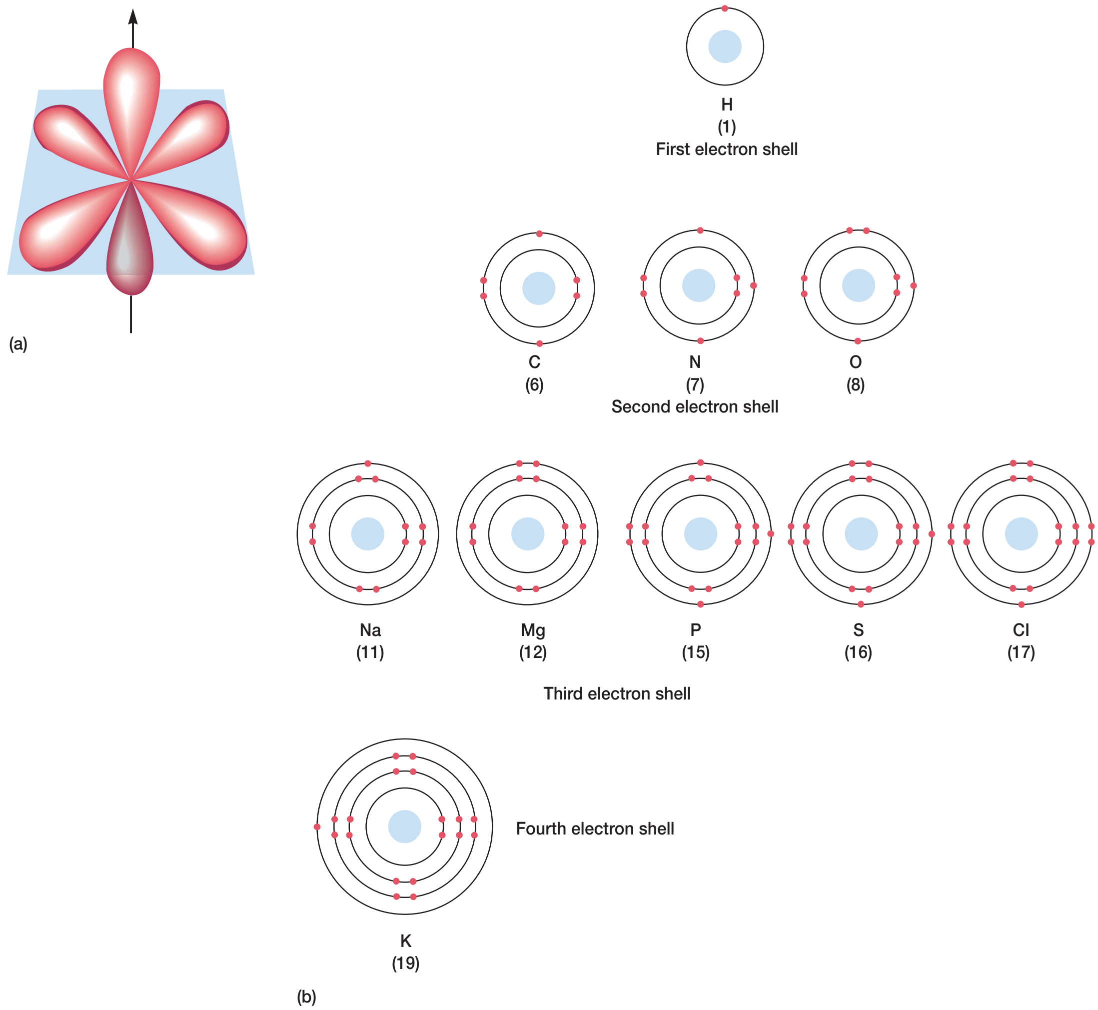


Figure AI.2 **Electron Orbitals.** (**a**) The three dumbbell-shaped orbitals of the second shell. The orbitals lie at right angles to each other. (**b**) The distribution of electrons in some common elements. Atomic numbers are given in parentheses.

and moving outward. For example, carbon has six electrons, two in its first shell and four in the second (figures AI.1 and AI.2*b*). The electrons in the outermost shell are the ones that participate in chemical reactions. The most stable condition is achieved when the outer shell is filled with electrons. Thus the number of bonds an element can form depends on the number of electrons required to fill the outer shell. Since carbon has four electrons in its outer shell and the shell is filled when it contains eight electrons, it can form four covalent bonds (table AI.1).

Chemical Bonds

Molecules are formed when two or more atoms associate through chemical bonding. Chemical bonds are attractive forces that hold together atoms, ions, or groups of atoms in a molecule or other substance. Many types of chemical bonds are present in organic molecules; three of the most important are covalent bonds, ionic bonds, and hydrogen bonds.

In covalent bonds, atoms are joined together by sharing pairs of electrons (**figure AI.3**). If the electrons are equally shared between identical atoms (e.g., in a carbon-carbon bond), the covalent bond is strong and nonpolar. When two different atoms such as carbon and oxygen share electrons, the covalent bond formed is polar since the electrons are pulled toward the more electronegative atom, the atom that more strongly attracts electrons. A single pair of electrons is shared in a single bond; a double bond is formed when two pairs of electrons are shared.

Atoms often contain either more or fewer electrons than the number of protons in their nuclei. When this is the case, they carry a net negative or positive charge and are called ions. Cations carry positive

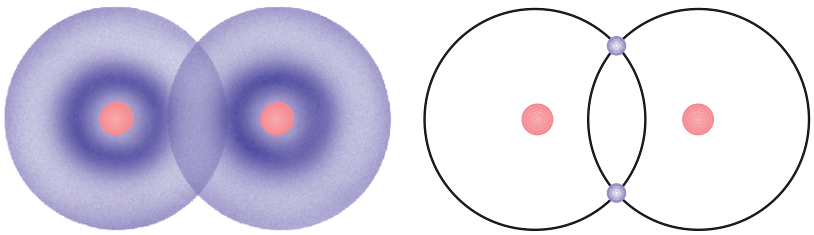


Figure AI.3 The Covalent Bond. A hydrogen molecule is formed when two hydrogen atoms share electrons.

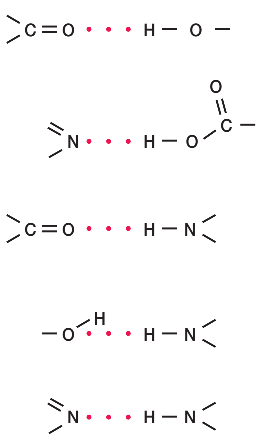


Figure AI.4 Hydrogen Bonds. Representative examples of hydrogen bonds present in biological molecules.

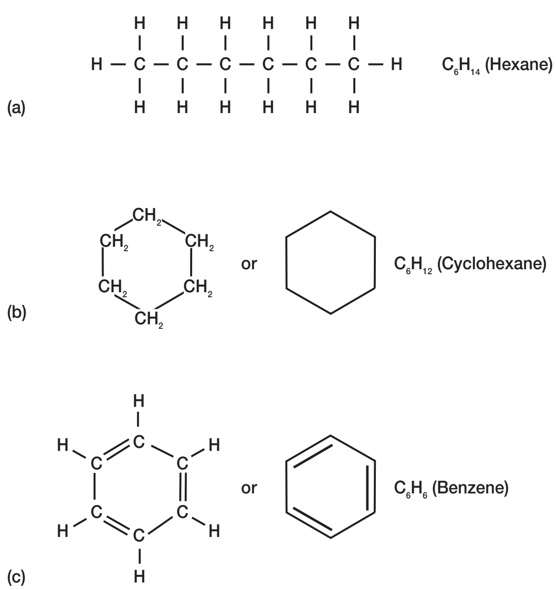


Figure AI.5 Hydrocarbons. Examples of hydrocarbons that are (**a**) linear, (**b**) cyclic, and (**c**) aromatic.

charges and anions have a net negative charge. When a cation and an anion approach each other, they are attracted by their opposite charges. This ionic attraction that holds two groups together is called an ionic bond. Ionic bonds are much weaker than covalent bonds and are easily disrupted by a polar solvent such as water. For example, the Na^+ cation is strongly attracted to the Cl^- anion in a sodium chloride crystal, but sodium chloride dissociates into separate ions (ionizes) when dissolved in water. Ionic bonds are important in the structure and function of proteins and other biological molecules.

When a hydrogen atom is covalently bonded to a more electronegative atom such as oxygen or nitrogen, the electrons are unequally shared and the hydrogen atom carries a partial positive charge. It will be attracted to an electronegative atom such as oxygen or nitrogen, which carries an unshared pair of electrons; this attraction is called a hydrogen bond (**figure AI.4**). Although an individual hydrogen bond is weak, there are so many hydrogen bonds in proteins and nucleic acids that they play a major role in determining protein and nucleic acid structure.

Organic Molecules

Most molecules in cells are organic molecules, molecules that contain carbon. Since carbon has four electrons in its outer shell, it tends to form four covalent bonds in order to fill its outer shell with eight electrons. This property makes it possible to form chains and rings of carbon atoms that also can bond with hydrogen and other atoms (**figure AI.5**). Although adjacent carbons usually are connected by single bonds, they may be joined by double or triple bonds. Rings that have alternating single and double bonds, like the benzene ring, are called aromatic rings. The hydrocarbon chain or ring provides a chemically inactive skeleton to which more reactive groups of atoms may be attached. These reactive groups with specific properties are known as functional groups. They usually contain atoms of oxygen, nitrogen, phosphorus, or sulfur (**figure AI.6**) and are largely responsible for most characteristic chemical properties of organic molecules.

Organic molecules are often divided into classes based on the nature of their functional groups. Ketones have a carbonyl group within the carbon chain, whereas alcohols have a hydroxyl on the chain. Organic acids have a carboxyl group, and amines have an amino group (**figure AI.7**).

Organic molecules may have the same chemical composition and yet differ in their molecular structure and properties. Such molecules are called isomers. One important class of isomers is the stereoisomers. Stereoisomers have the same atoms arranged in the same nucleus-to-nucleus sequence but differ in the spatial arrangement of their atoms. For example, an amino acid such as alanine can form stereoisomers

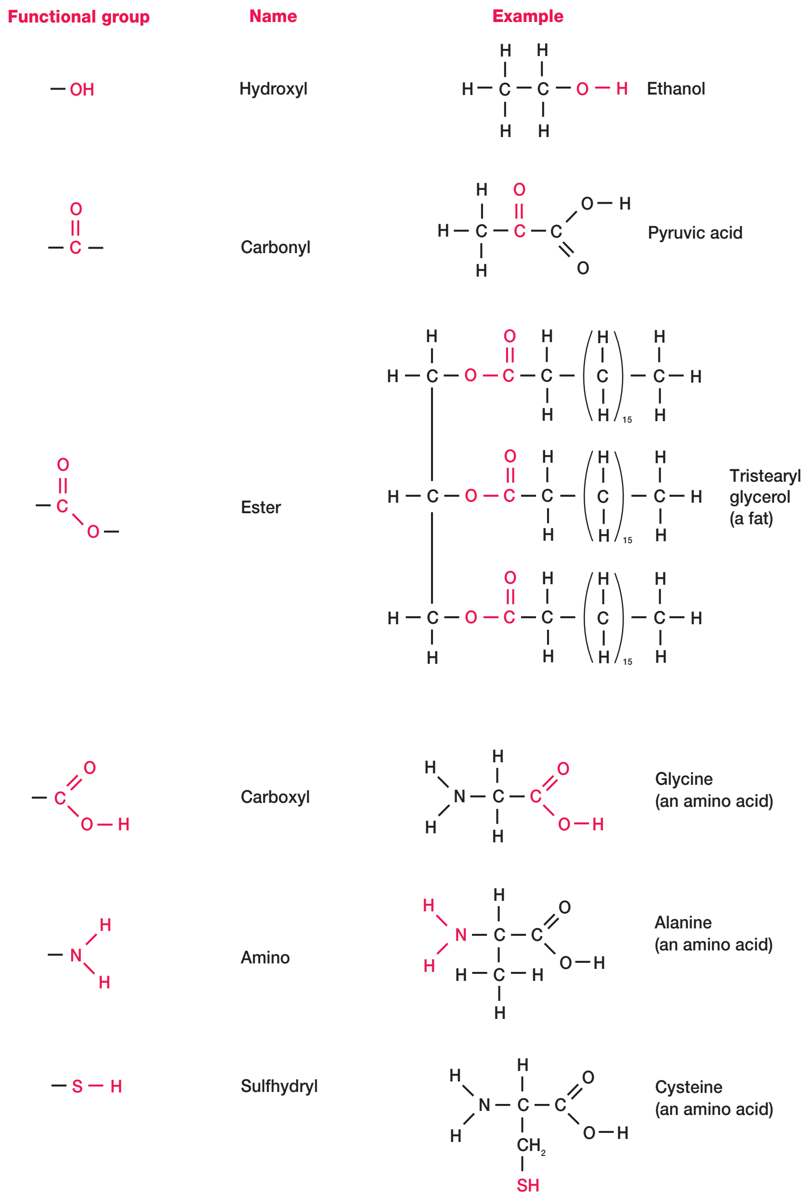


Figure AI.6 Functional Groups. Some common functional groups in organic molecules. The groups are shown in color.

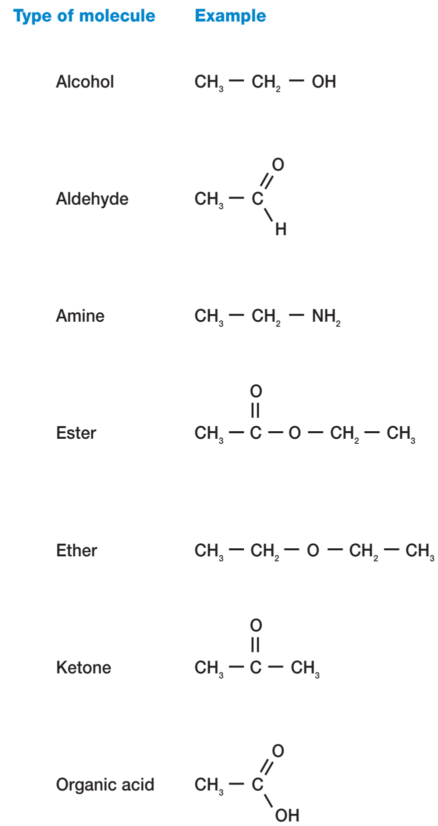


Figure AI.7 Types of Organic Molecules. These are classified on the basis of their functional groups.

(**figure AI.8**). L-Alanine and other L-amino acids are the stereoisomer forms normally present in proteins.

Carbohydrates

Carbohydrates are aldehyde or ketone derivatives of polyhydroxy alcohols. The smallest and least complex carbohydrates are the simple sugars or monosaccharides. The most common sugars have five or six carbons (**figure AI.9**). A sugar in its ring form has two isomeric structures, the α and β forms, that differ in the orientation of the hydroxyl on the aldehyde or ketone carbon, which is called the anomeric or glycosidic carbon (**figure AI.10**). Microorganisms have many sugar derivatives in which a hydroxyl is replaced by an amino group or some other functional group (e.g., glucosamine).

Two monosaccharides can be joined by a bond between the anomeric carbon of one sugar and a hydroxyl or the anomeric carbon of the second (**figure AI.11**). The bond joining sugars is a glycosidic bond and may be either α or β depending on the orientation of the anomeric carbon. Two sugars linked in this way constitute a disaccharide. Some

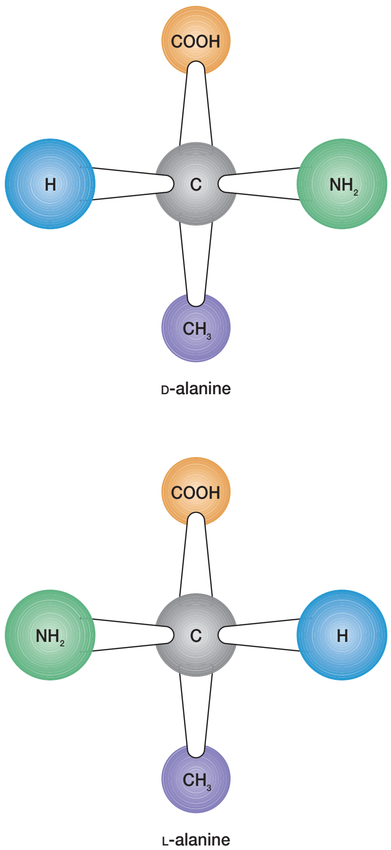


Figure AI.8 The Stereoisomers of Alanine. The α-carbon is in gray, L-alanine is the form usually present in proteins.

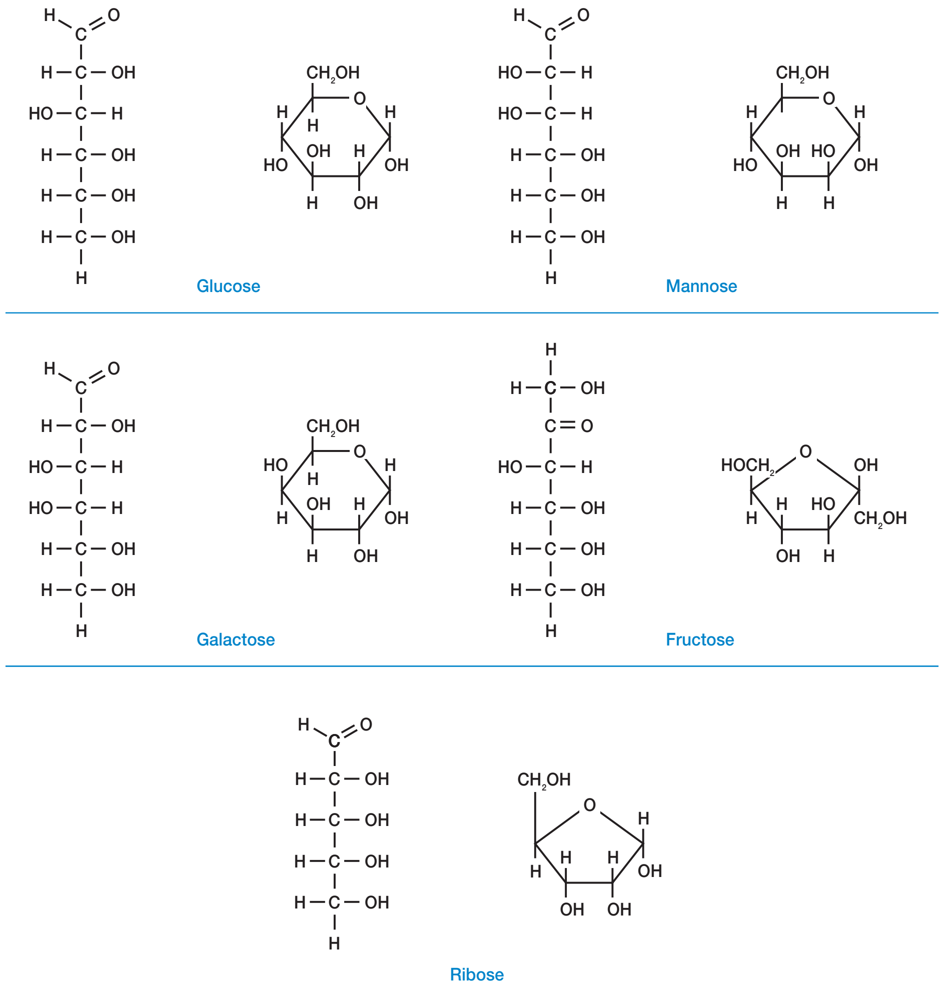


Figure AI.9 Common Monosaccharides. Structural formulas for both the open chains and the ring forms are provided.

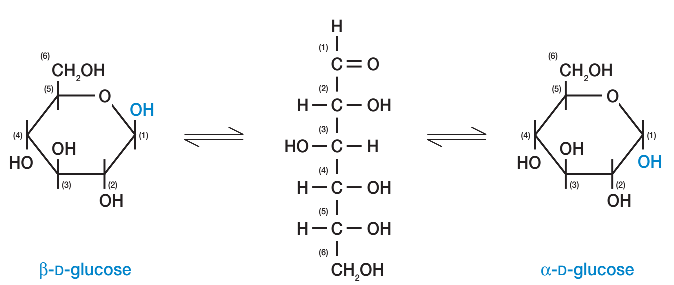


Figure AI.10 The Interconversion of Monosaccharide Structures. The open chain form of glucose and other sugars is in equilibrium with closed ring structures (depicted here with Haworth projections). Aldehyde sugars form cyclic hemiacetals, and keto sugars produce cyclic hemiketals. When the hydroxyl on carbon one of cyclic hemiacetals projects above the ring, the form is known as a β form. The α form has a hydroxyl that lies below the plane of the ring. The same convention is used in showing the α and β forms of hemiketals such as those formed by fructose.

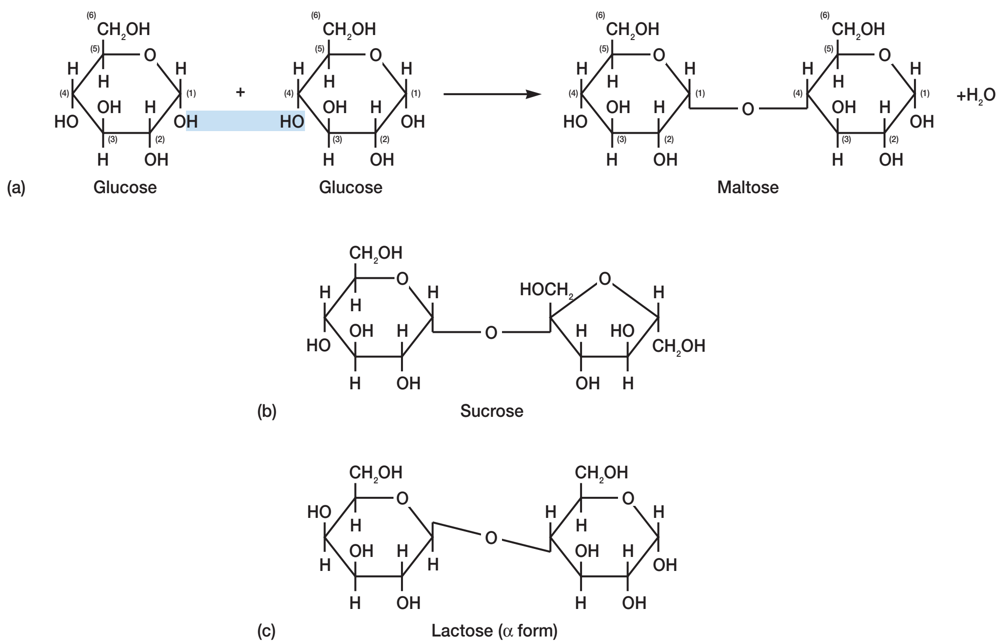


Figure AI.11 Common Disaccharides. (**a**) The formation of maltose from two molecules of an α-glucose. The bond connecting the glucose extends between carbons one and four, and involves the α form of the anomeric carbon. Therefore, it is called an α (1 → 4) glycosidic bond. (**b**) Sucrose is composed of a glucose and a fructose joined to each other through their anomeric carbons, and αβ (1 → 2) bond. (**c**) The milk sugar lactose contains galactose and glucose joined by a β (1 → 4) glycosidic bond.

common disaccharides are maltose (two glucose molecules), lactose (glucose and galactose), and sucrose (glucose and fructose). If 10 or more sugars are linked together by glycosidic bonds, a polysaccharide is formed. For example, starch and glycogen are common polymers of glucose that are used as sources of carbon and energy (**figure AI.12**).

Lipids

All cells contain a heterogeneous mixture of organic molecules that are relatively insoluble in water but very soluble in nonpolar solvents such as chloroform, ether, and benzene. These molecules are called lipids. Lipids vary greatly in structure and include triacylglycerols, phospholipids, steroids, carotenoids, and many other types. Among other functions, they serve as membrane components, storage forms for carbon and energy, precursors of other cell constituents, and protective barriers against water loss.

Most lipids contain fatty acids, monocarboxylic acids that often are straight chained but may be branched. Saturated fatty acids lack double bonds in their carbon chains, whereas unsaturated fatty acids have double bonds. The most common fatty acids are 16 or 18 carbons long.

Two good examples of common lipids are triacylglycerols and phospholipids. Triacylglycerols are composed of glycerol esterified to three fatty acids (**figure AI.13*a***). They are used to store carbon and energy. Phospholipids are lipids that contain at least one phosphate group and often have a nitrogenous constituent as well. Phosphatidyl ethanolamine is an important phospholipid frequently present in bacterial membranes (figure AI.13*b*). It is composed of two fatty acids esterified to glycerol. The third glycerol hydroxyl is joined with a phosphate group, and ethanolamine is attached to the phosphate. The resulting lipid is very asymmetric with a hydrophobic nonpolar end contributed by the fatty acids and a polar, hydrophilic end. In cell membranes the hydrophobic end is buried in the interior of the membrane, while the polar-charged end is at the membrane surface and exposed to water.

Proteins

The basic building blocks of proteins are amino acids. An amino acid contains a carboxyl group and an amino group on its alpha carbon (**figure AI.14**). About 20 amino acids are normally found in proteins; they differ from each other with respect to their side chains (**figure AI.15**). In proteins, amino acids are linked together by peptide bonds between their carboxyls and α-amino groups to form linear polymers called peptides (**figure AI.16**). If a peptide contains more than 10 amino acids, it usually is called a polypeptide. Each protein is composed of one or more polypeptide chains and has a molecular weight greater than about 6,000 to 7,000.

Proteins have three or four levels of structural organization and complexity. The primary structure of a protein is the sequence of the amino acids in its polypeptide chain or chains. The structure of the polypeptide chain backbone is also considered part of the primary structure. Each different polypeptide has its own amino acid sequence that is a reflection of the nucleotide sequence in the gene that codes for its synthesis. The polypeptide chain can coil along one axis in space into various shapes like the α-helix

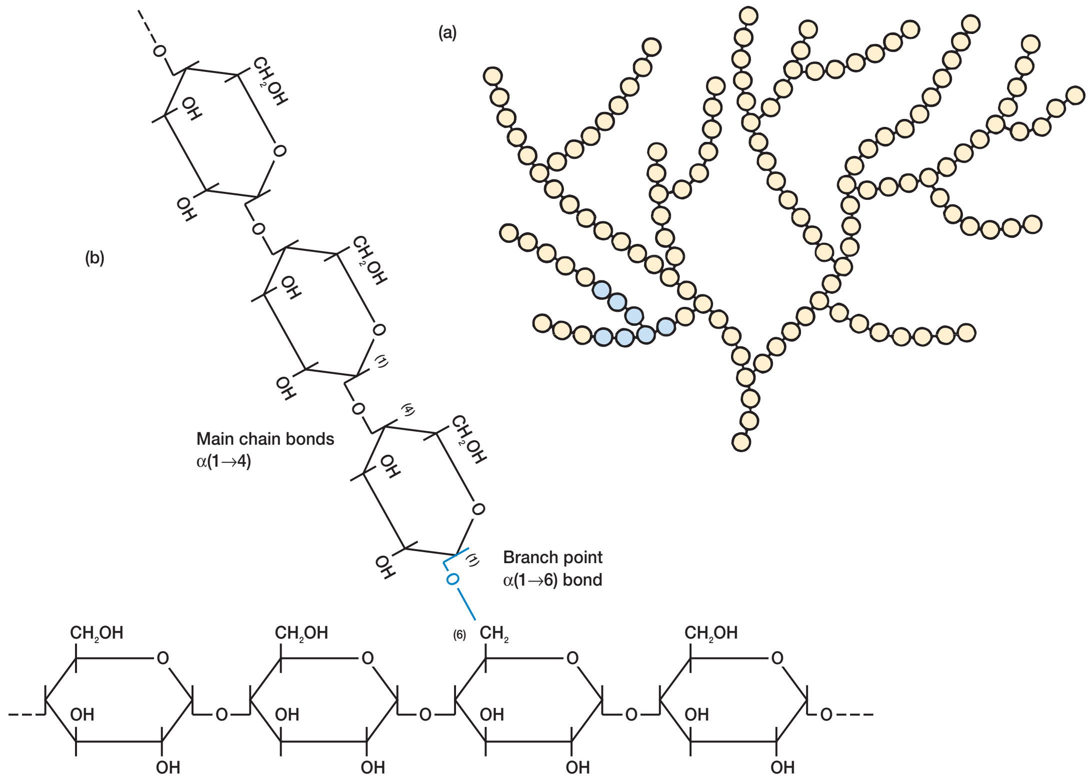


Figure AI.12 Glycogen and Starch Structure. (**a**) An overall view of the highly branched structure characteristic of glycogen and most starch. The circles represent glucose residues. (**b**) A close-up of a small part of the chain (shown in color in part ***a***) revealing a branch point with its α (1 → 6) glycosidic bond, which is colored blue.

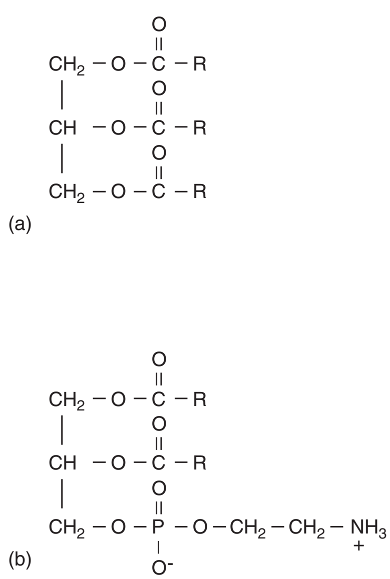


Figure AI.13 Examples of Common Lipids. (**a**) A triacylglycerol or neutral fat. (**b**) The phospholipid phosphatidyl ethanolamine. The R groups represent fatty acid side chains.

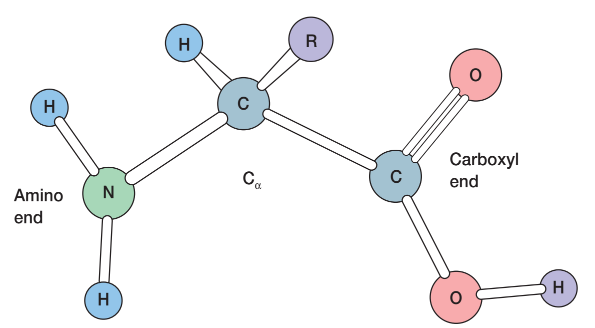


Figure AI.14 L-Amino Acid Structure. The uncharged form is shown.

Figure AI.15 The Common Amino Acids. The structures of the α-amino acids normally found in proteins. Their side chains are shown in color, and they are grouped together based on the nature of their side chains—nonpolar, polar, negatively charged (acid), or positively charged (basic). Proline is actually an imino acid rather than an amino acid.

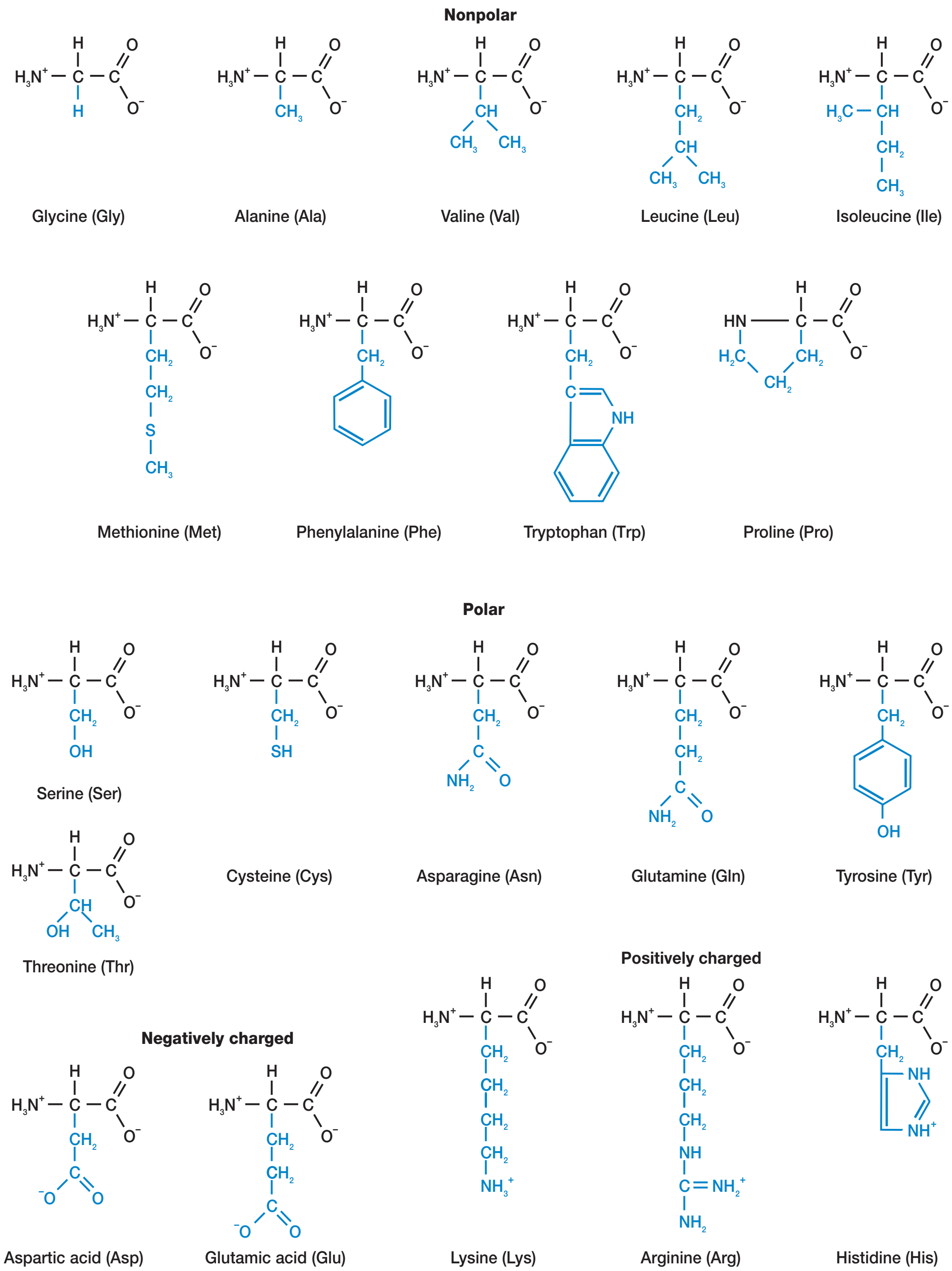
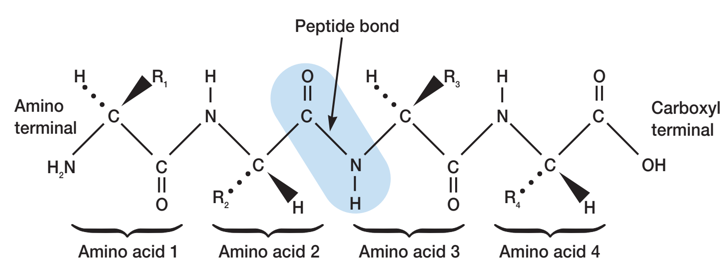


Figure AI.16 A Tetrapeptide Chain. The end of the chain with a free α-amino group is the amino or N terminal. The end with the free α-carboxyl is the carboxyl or C terminal. One peptide bond is shown in color.



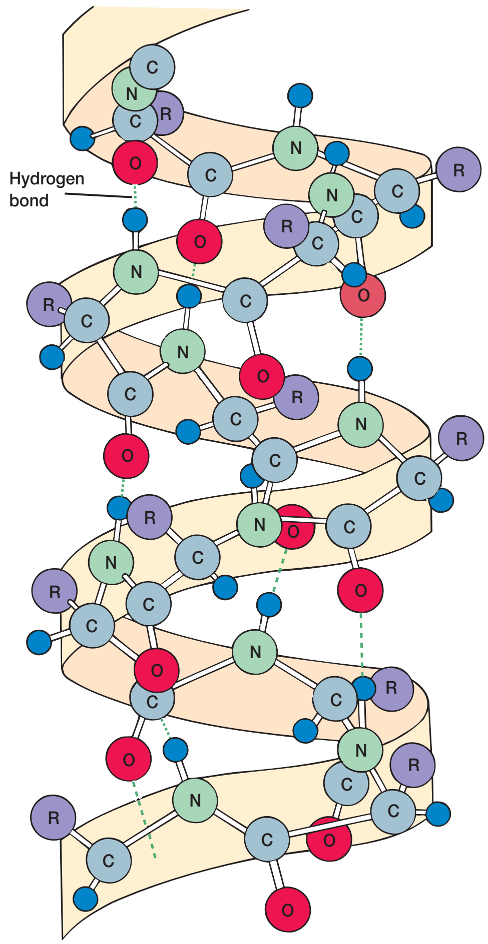


Figure AI.17 The α-Helix. A polypeptide twisted into one type of secondary structure, the α-helix. The helix is stabilized by hydrogen bonds joining peptide bonds that are separated by three amino acids.

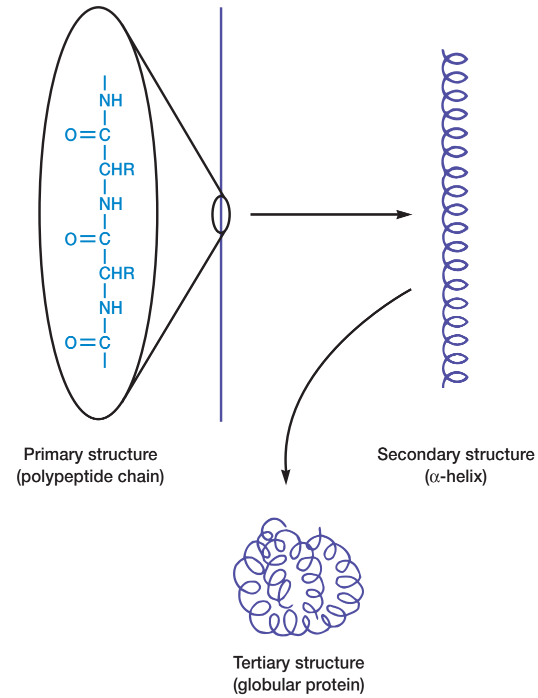


Figure AI.18 Secondary and Tertiary Protein Structures. The formation of secondary and tertiary protein structures by folding a polypeptide chain with its primary structure.

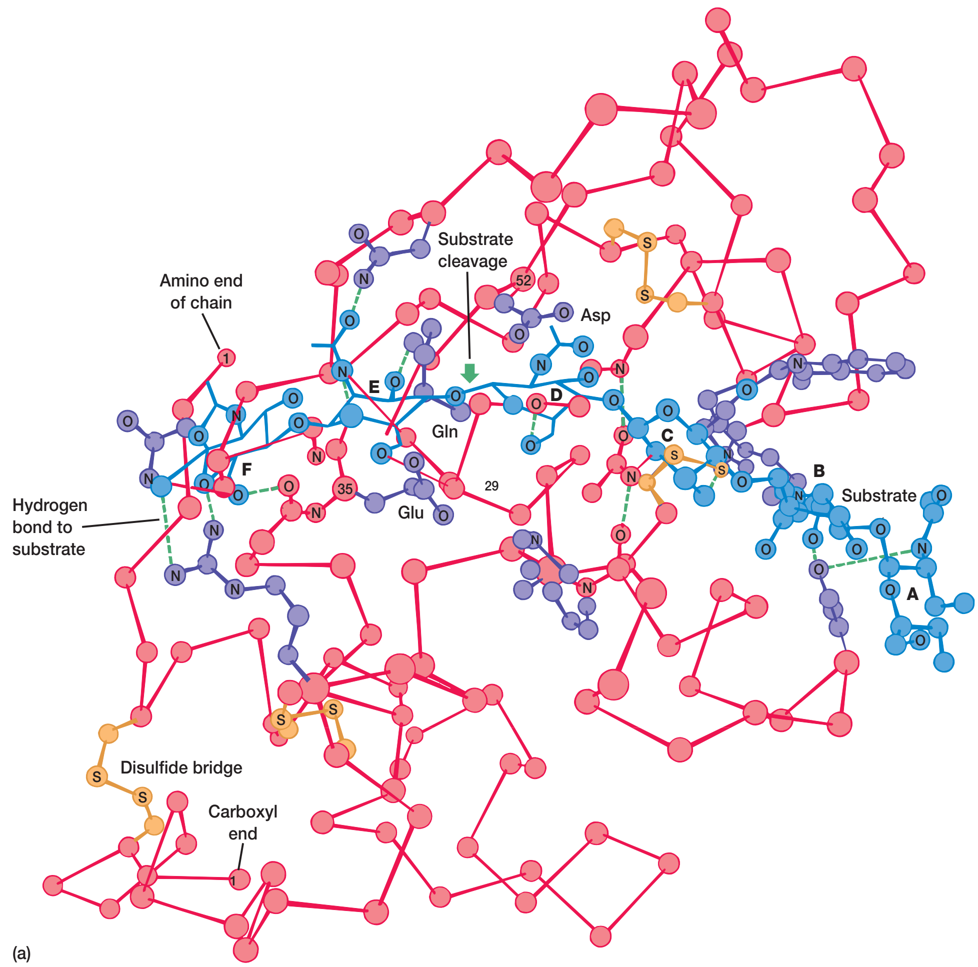
(**figure AI.17**). This arrangement of the polypeptide in space around a single axis is called the secondary structure. Secondary structure is formed and stabilized by the interactions of amino acids that are fairly close to one another on the polypeptide chain. The polypeptide with its primary and secondary structure can be coiled or organized in space along three axes to form a more complex, three-dimensional shape (**figure AI.18**). This level of organization is the tertiary structure (**figure AI.19**). Amino acids more distant from one another on the polypeptide chain contribute to tertiary structure. Secondary and tertiary structures are examples of conformation, molecular shape that can be changed by bond rotation and without breaking covalent bonds. When a protein contains more than one polypeptide chain, each chain with its own primary, secondary, and tertiary structure associates with the other chains to form the final molecule. The way in which polypeptides associate with each other in space to form the final protein is called the protein's quaternary structure (**figure AI.20**).

The final conformation of a protein is ultimately determined by the amino acid sequence of its polypeptide chains. Under proper conditions a completely unfolded polypeptide will fold into its normal final shape without assistance.

Protein secondary, tertiary, and quaternary structure is largely determined and stabilized by many weak noncovalent forces such as hydrogen bonds and ionic bonds. Because of this, protein shape often is very flexible and easily changed. This flexibility is very important in protein function and in the regulation of enzyme activity. Because of their flexibility, however, proteins readily lose their proper shape and activity when exposed to harsh conditions. The only covalent bond commonly involved in the secondary and tertiary structure of proteins is the disulfide bond. The disulfide bond is formed when two cysteines are linked through their sulfhydryl groups. Disulfide bonds generally strengthen or stabilize protein structure but are not especially important in directly determining protein conformation.

Nucleic Acids

The nucleic acids, deoxyribonucleic acid (DNA) and ribonucleic acid (RNA), are polymers of deoxyribonucleosides and ribonucleosides joined by phosphate groups. The nucleosides in DNA contain the purines adenine and guanine, and the pyrimidine bases thymine and cytosine. In RNA the pyrimidine uracil is substituted for thymine. Because of their importance for genetics and molecular biology, the chemistry of nucleic acids is introduced earlier in the text. The structure and synthesis of purines and pyrimidines are discussed in chapter 10 (*pp. 217–18*). The structures of DNA and RNA are described in chapter 11 (*pp. 230–35*).



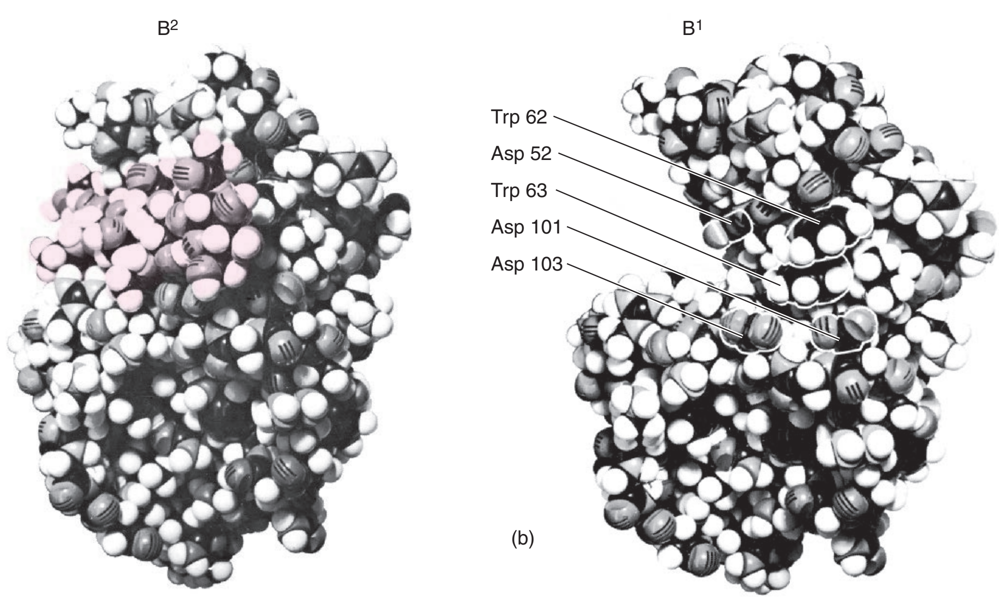


Figure AI.19 **Lysozyme.** The tertiary structure of the enzyme lysozyme. (**a**) A diagram of the protein's polypeptide backbone with the substrate hexasaccharide shown in blue. The point of substrate cleavage is indicated. (**b**) A space-filling model of lysozyme. The left figure shows the empty active site with some of its more important amino acids indicated. On the right the enzyme has bound its substrate (in pink).

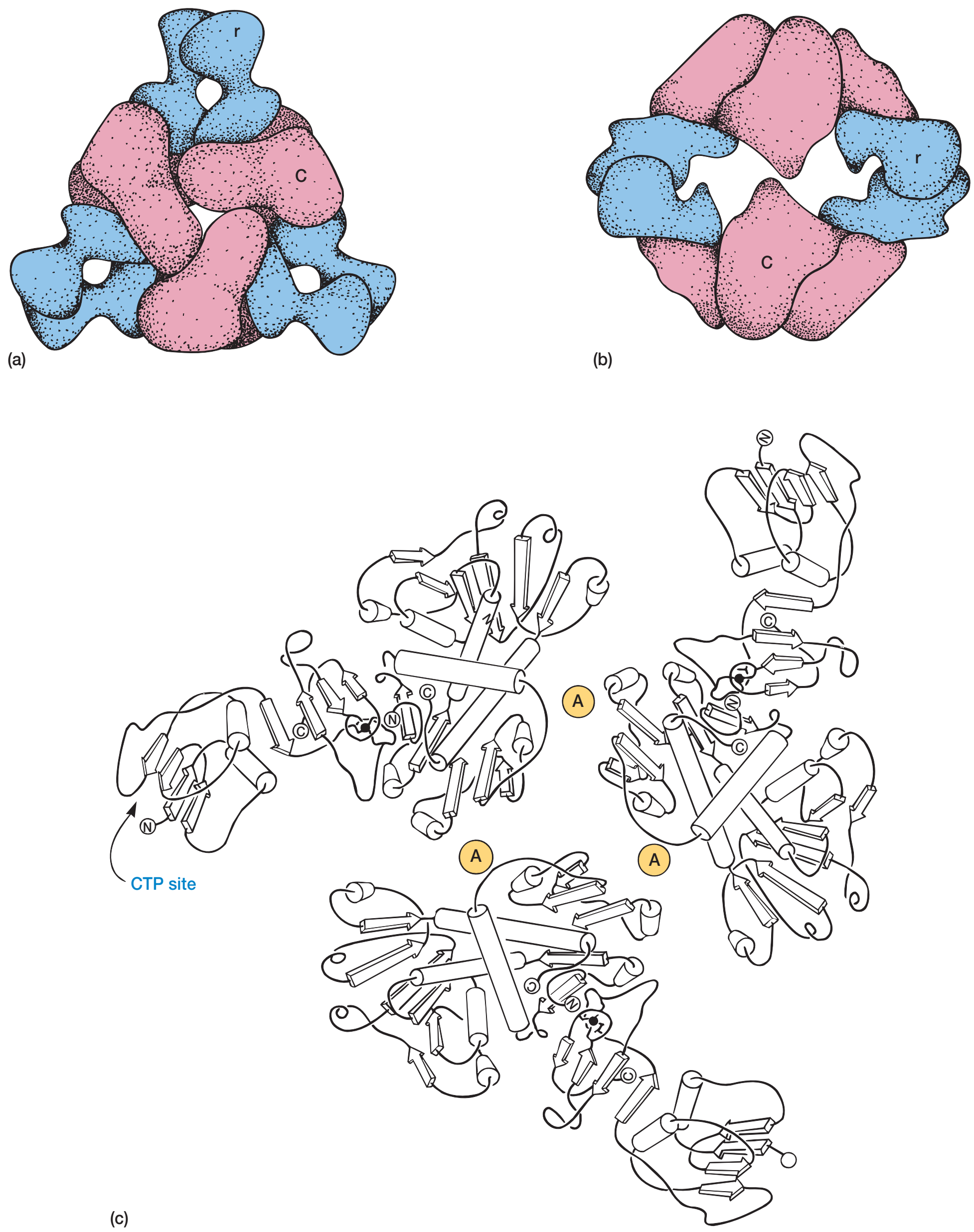


Figure AI.20 An Example of Quaternary Structure. The enzyme aspartate carbamoyltransferase from *E. coli* has two types of subunits, catalytic and regulatory. The association between the two types of subunits is shown: (**a**) a top view, and (**b**) a side view of the enzyme. The catalytic (C) and regulator (r) subunits are shown in different colors. (**c**) The peptide chains shown when viewed from the top as in (**a**). The active sites of the enzyme are located at the positions indicated by A. (*See pp. 166–68 for more details.*) (**a** and **b**) Adapted from Krause, et al., in *Proceedings of the National Academy of Sciences,* V. 82, 1985, as appeared in *Biochemistry,* 3rd edition by Lubert Stryer. Copyright © 1975, 1981, 1988. Reprinted with permission of W.H. Freeman and Company. (**c**) Adapted from Kantrowitz, et al., in *Trends in Biochemical Science,* V. 5, 1980, as appeared in *Biochemistry,* 3rd edition by Lubert Stryer. Copyright © 1975, 1981, 1988. Reprinted with permission of W.H. Freeman and Company.

Appendix II

Common Metabolic Pathways

This appendix contains a few of the more important pathways discussed in the text, particularly those involved in carbohydrate catabolism. Enzyme names and final end products are given in color. Consult the text for a description of each pathway and its roles.

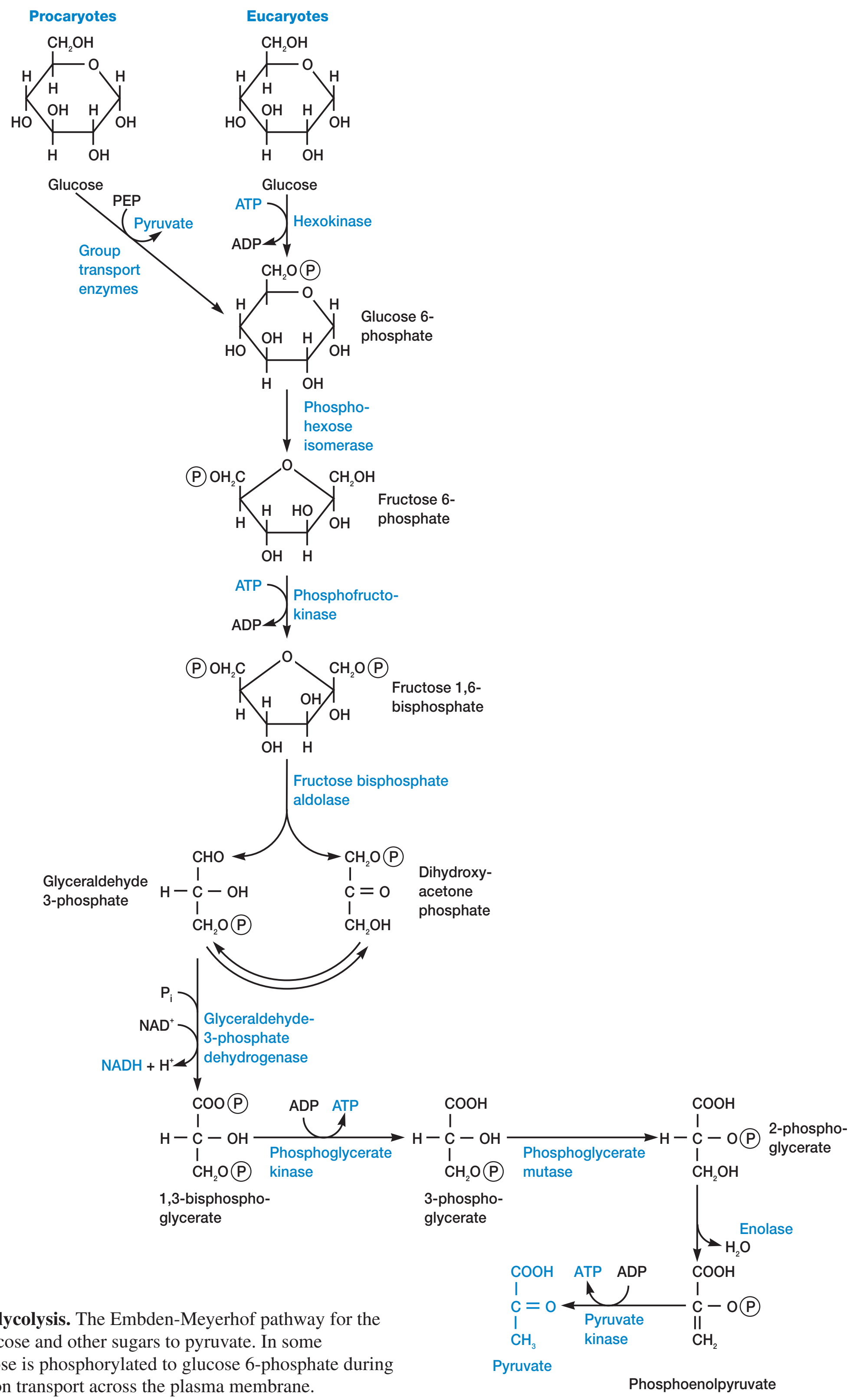


Figure AII.1 Glycolysis. The Embden-Meyerhof pathway for the conversion of glucose and other sugars to pyruvate. In some prokaryotes glucose is phosphorylated to glucose 6-phosphate during group translocation transport across the plasma membrane.

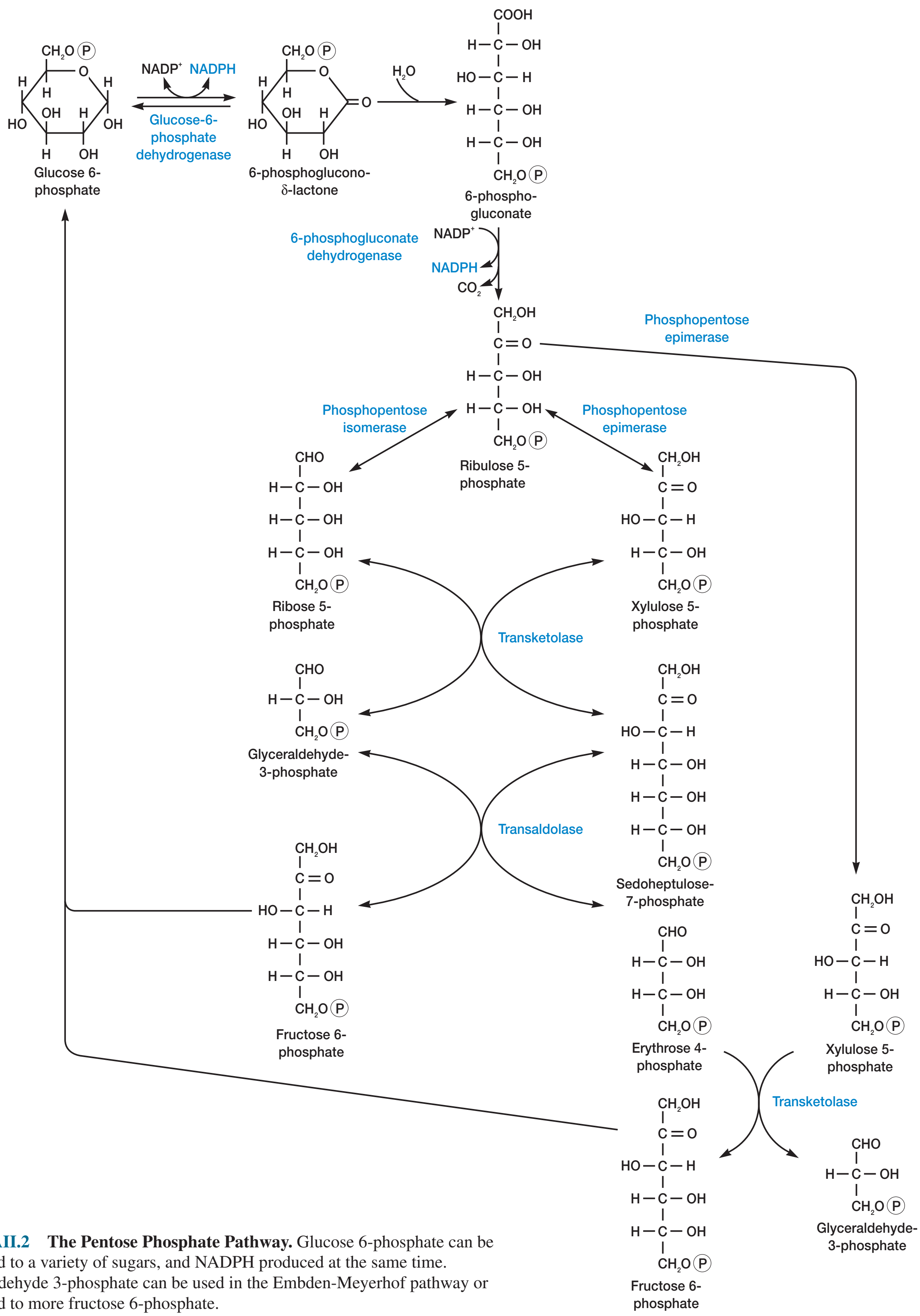


Figure AII.2 The Pentose Phosphate Pathway. Glucose 6-phosphate can be converted to a variety of sugars, and NADPH produced at the same time. Glyceraldehyde 3-phosphate can be used in the Embden-Meyerhof pathway or converted to more fructose 6-phosphate.

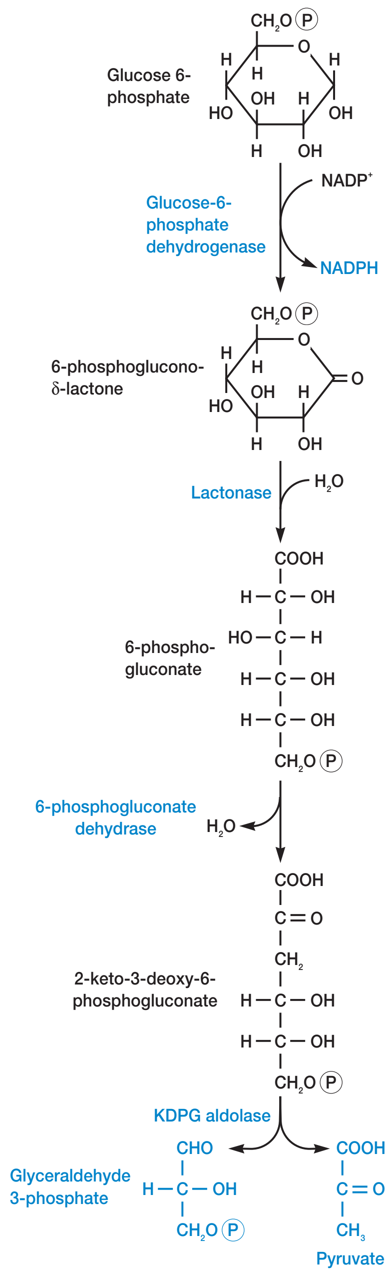


Figure AII.3 **The Entner-Doudoroff Pathway.**

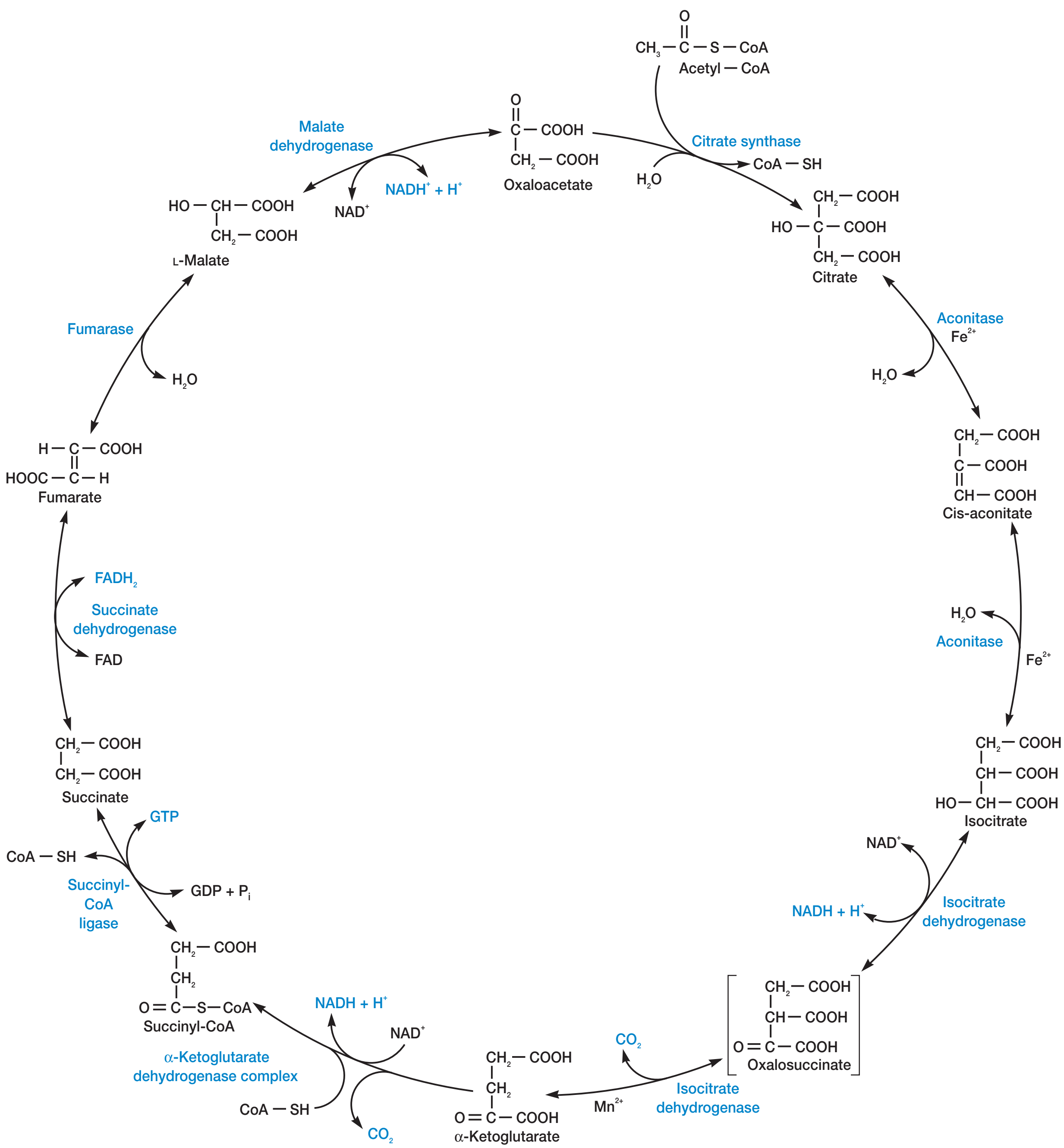


Figure AII.4 The Tricarboxylic Acid Cycle. Cis-aconitate and oxalosuccinate remain bound to aconitase and isocitrate dehydrogenase. Oxalosuccinate has been placed in brackets since it is so unstable.

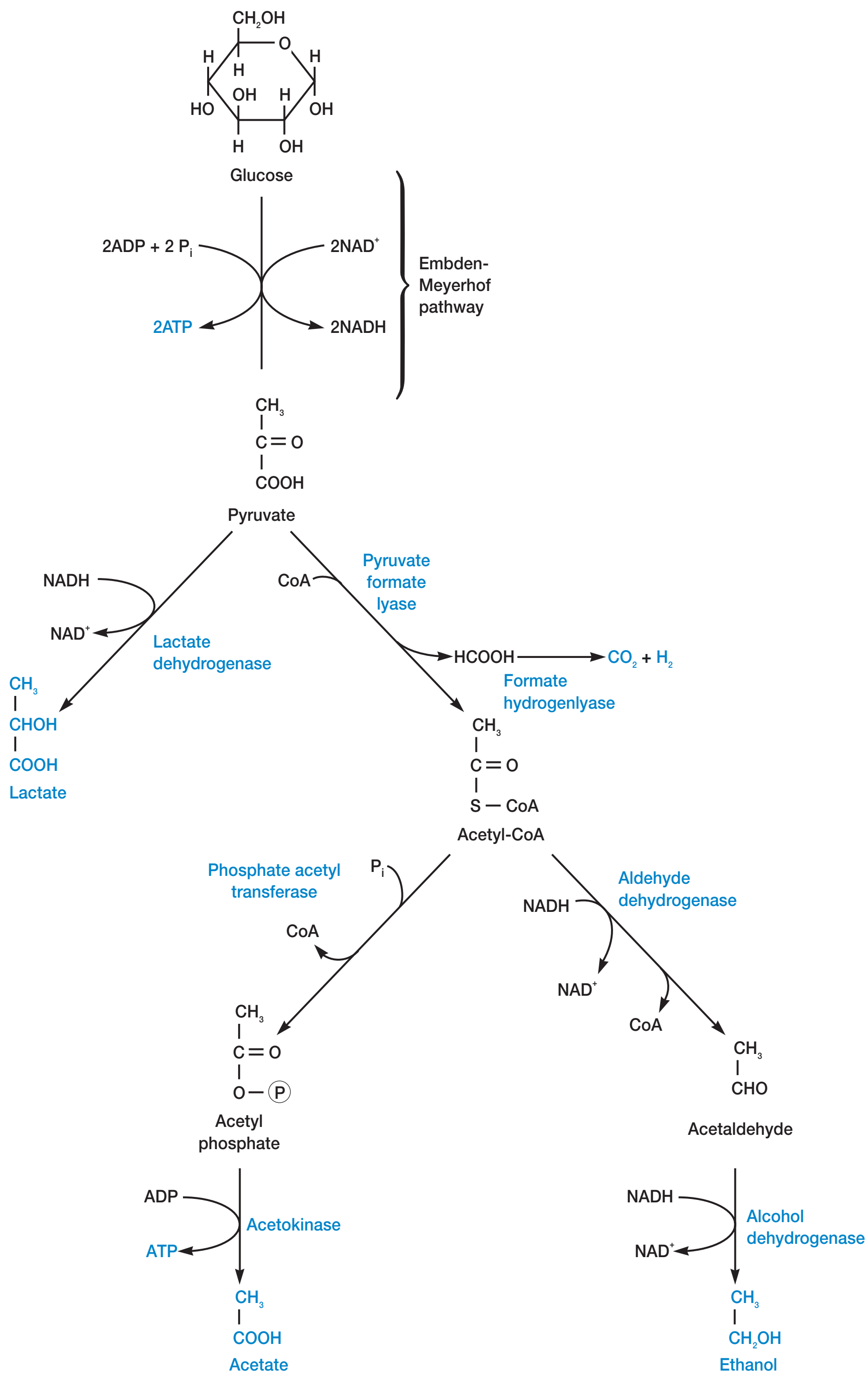


Figure AII.5 The Mixed Acid Fermentation Pathway. This pathway is characteristic of many members of the *Enterobacteriaceae* such as *E. coli.*

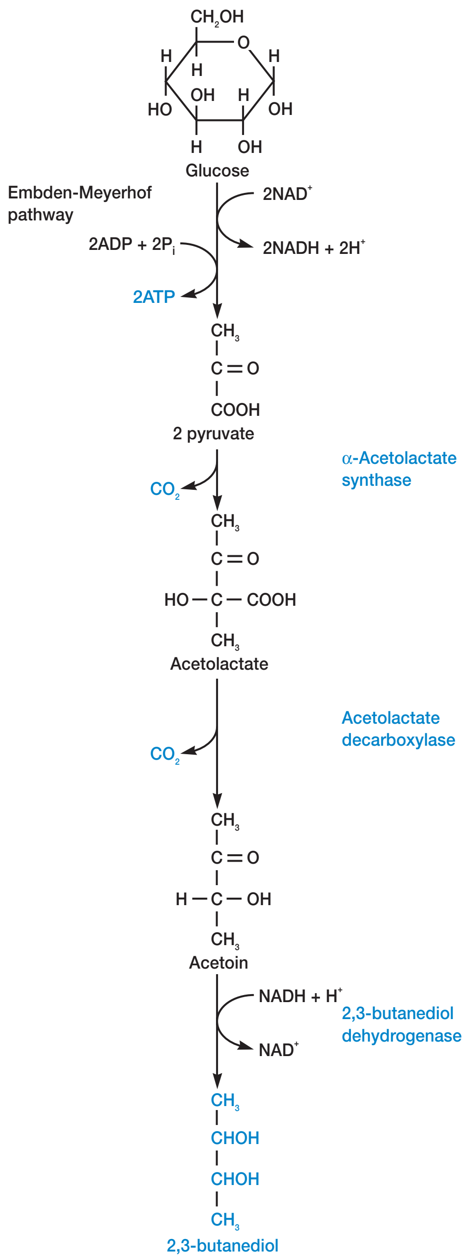
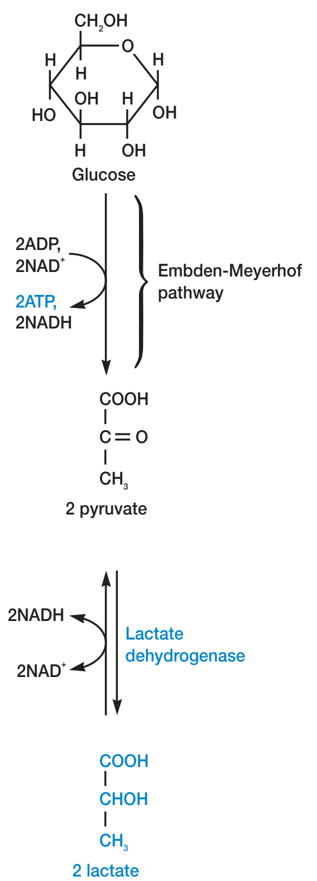
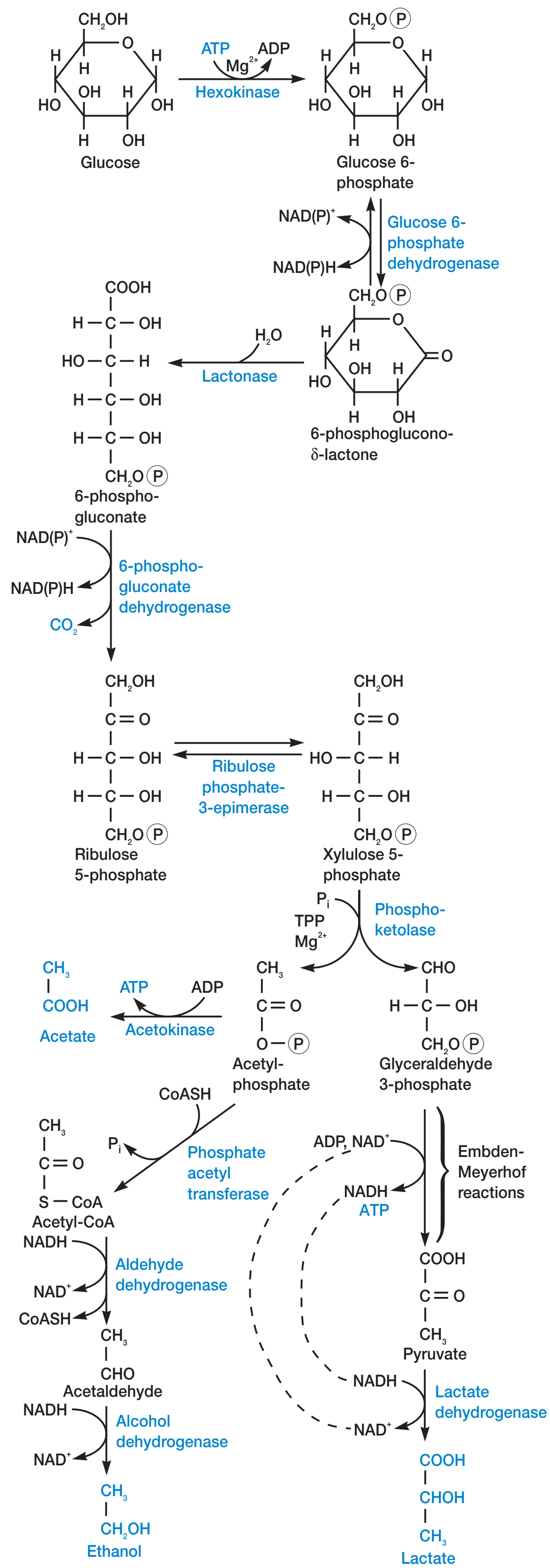


Figure AII.6 The Butanediol Fermentation Pathway. This pathway is characteristic of members of the *Enterobacteriaceae* such as *Enterobacter.* Other products may also be formed during butanediol fermentation.



(a)



(b)

Figure AII.7 Lactic Acid Fermentations.
(**a**) Homolactic fermentation pathway. (**b**) Heterolactic fermentation pathway.

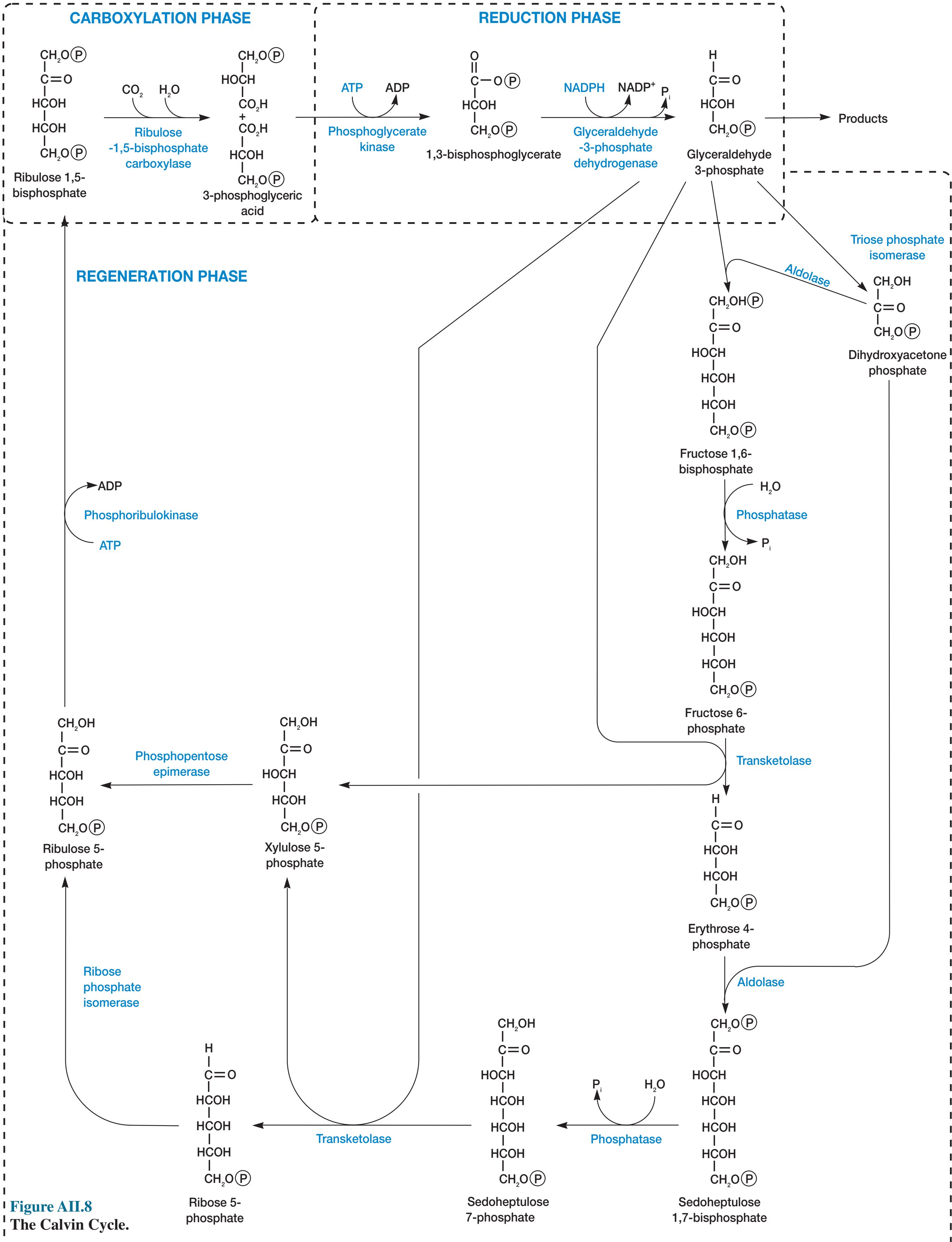


Figure AII.8
The Calvin Cycle.

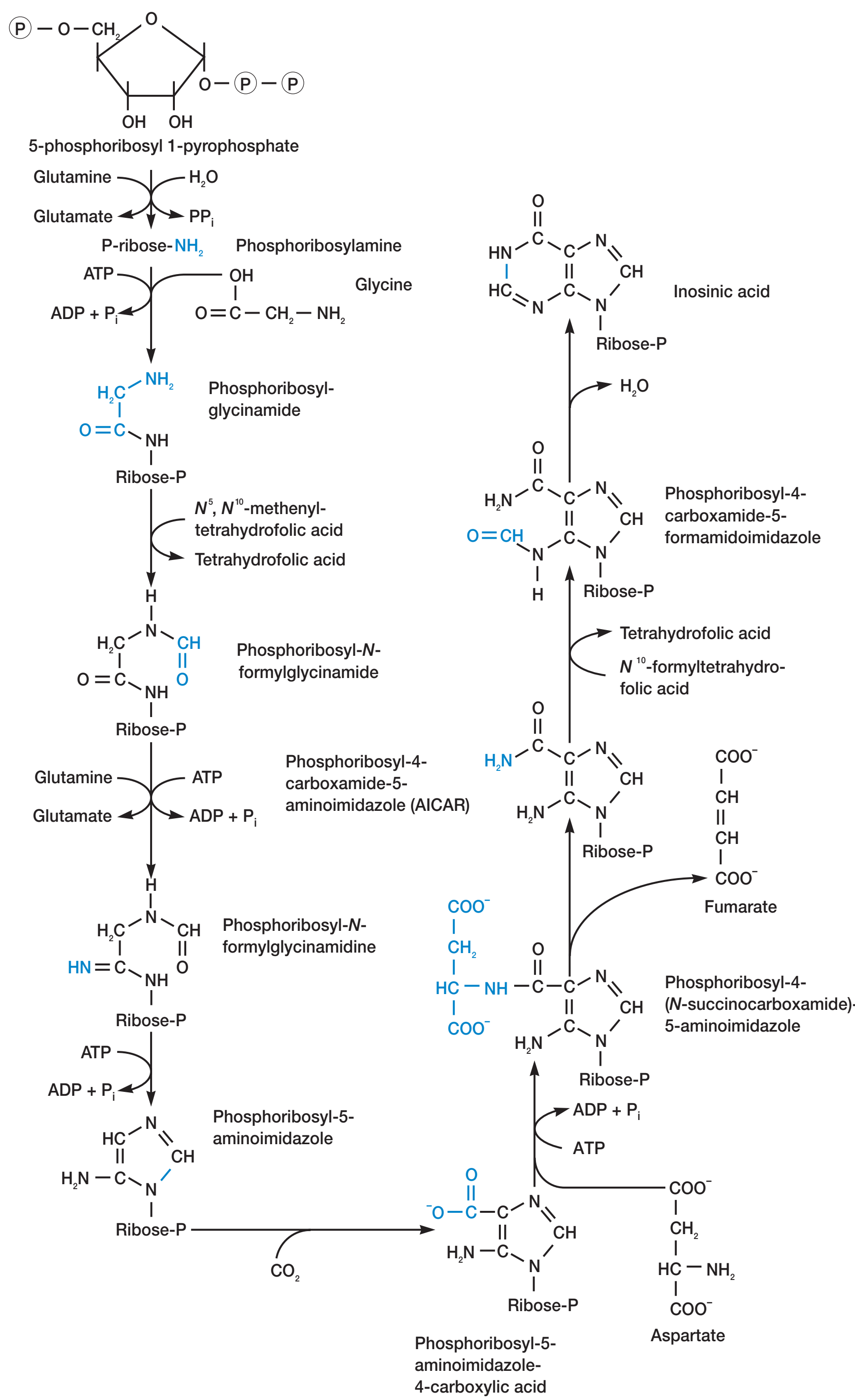


Figure AII.9 The Pathway for Purine Biosynthesis. Inosinic acid is the first purine end product. The purine skeleton is constructed while attached to a ribose phosphate.

APPENDIX III

Classification of Procaryotes According to the First Edition of *Bergey's Manual of Systematic Bacteriology*

Great progress has occurred in bacterial taxonomy since the first edition of *Bergey's Manual of Systematic Bacteriology* was published between 1984 and 1989. Many completely new taxa have been added, and large, complex genera such as *Pseudomonas, Streptococcus,* and *Bacillus* have been subdivided to form new genera. Thus the classification presented here differs from that given in the 1994 edition of *Bergey's Manual of Determinative Bacteriology,* and even this more recent work is already out-of-date. Appendix III summarizes the system presented in the first edition of *Bergey's Manual of Systematic Bacteriology.*

Volume I*

Section 1

The Spirochetes

Order I *Spirochaetales*
- Family I *Spirochaetaceae*
  - Genus I *Spirochaeta*
  - Genus II *Cristispira*
  - Genus III *Treponema*
  - Genus IV *Borrelia*
- Family II *Leptospiraceae*
  - Genus I *Leptospira*

Other Organisms
- Hindgut Spirochetes of Termites and *Cryptocercus punctulatus*

Section 2

Aerobic/Microaerophilic, Motile, Helical/Vibrioid Gram-Negative Bacteria

- Genus *Aquaspirillum*
- Genus *Spirillum*
- Genus *Azospirillum*
- Genus *Oceanospirillum*
- Genus *Campylobacter*
- Genus *Bdellovibrio*
- Genus *Vampirovibrio*

Section 3

Nonmotile (or Rarely Motile), Gram-Negative Curved Bacteria

- Family I *Spirosomaceae*
  - Genus I *Spirosoma*
  - Genus II *Runella*
  - Genus III *Flectobacillus*
- Other Genera
  - Genus *Microcyclus*
  - Genus *Meniscus*
  - Genus *Brachyarcus*
  - Genus *Pelosigma*

Section 4

Gram-Negative Aerobic Rods and Cocci

- Family I *Pseudomonadaceae*
  - Genus I *Pseudomonas*
  - Genus II *Xanthomonas*
  - Genus III *Frateuria*
  - Genus IV *Zoogloea*
- Family II *Azotobacteraceae*
  - Genus I *Azotobacter*
  - Genus II *Azomonas*
- Family III *Rhizobiaceae*
  - Genus I *Rhizobium*
  - Genus II *Bradyrhizobium*
  - Genus III *Agrobacterium*
  - Genus IV *Phyllobacterium*
- Family IV *Methylococcaceae*
  - Genus I *Methylococcus*
  - Genus II *Methylomonas*
- Family V *Halobacteriaceae*
  - Genus I *Halobacterium*
  - Genus II *Halococcus*
- Family VI *Acetobacteraceae*
  - Genus I *Acetobacter*
  - Genus II *Gluconobacter*
- Family VII *Legionellaceae*
  - Genus I *Legionella*
- Family VIII *Neisseriaceae*
  - Genus I *Neisseria*
  - Genus II *Moraxella*
  - Genus III *Acinetobacter*
  - Genus IV *Kingella*
- Other Genera
  - Genus *Beijerinckia*
  - Genus *Derxia*
  - Genus *Xanthobacter*
  - Genus *Thermus*
  - Genus *Thermomicrobium*
  - Genus *Halomonas*
  - Genus *Alteromonas*
  - Genus *Flavobacterium*
  - Genus *Alcaligenes*
  - Genus *Serpens*
  - Genus *Janthinobacterium*
  - Genus *Brucella*
  - Genus *Bordetella*
  - Genus *Francisella*
  - Genus *Paracoccus*
  - Genus *Lampropedia*

Section 5

Facultatively Anaerobic Gram-Negative Rods

- Family I *Enterobacteriaceae*
  - Genus I *Escherichia*
  - Genus II *Shigella*
  - Genus III *Salmonella*
  - Genus IV *Citrobacter*
  - Genus V *Klebsiella*
  - Genus VI *Enterobacter*
  - Genus VII *Erwinia*
  - Genus VIII *Serratia*
  - Genus IX *Hafnia*
  - Genus X *Edwardsiella*
  - Genus XI *Proteus*
  - Genus XII *Providencia*
  - Genus XIII *Morganella*
  - Genus XIV *Yersina*
- Other Genera of the Family *Enterobacteriaceae*
  - Genus *Obesumbacterium*
  - Genus *Xenorhabdus*
  - Genus *Kluyvera*
  - Genus *Rahnella*
  - Genus *Cedecea*
  - Genus *Tatumella*
- Family II *Vibrionaceae*
  - Genus I *Vibrio*
  - Genus II *Photobacterium*
  - Genus III *Aeromonas*
  - Genus IV *Plesiomonas*

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Family III *Pasteurellaceae*
Genus I *Pasteurella*
Genus II *Haemophilus*
Genus III *Actinobacillus*
Other Genera
Genus *Zymomonas*
Genus *Chromobacterium*
Genus *Cardiobacterium*
Genus *Calymmatobacterium*
Genus *Gardnerella*
Genus *Eikenella*
Genus *Streptobacillus*

Section 6

Anaerobic Gram-Negative Straight, Curved, and Helical Rods

Family I *Bacteroidaceae*
Genus I *Bacteroides*
Genus II *Fusobacterium*
Genus III *Leptotrichia*
Genus IV *Butyrivibrio*
Genus V *Succinimonas*
Genus VI *Succinivibrio*
Genus VII *Anaerobiospirillum*
Genus VIII *Wolinella*
Genus IX *Selenomonas*
Genus X *Anaerovibrio*
Genus XI *Pectinatus*
Genus XII *Acetivibrio*
Genus XIII *Lachnospira*

Section 7

Dissimilatory Sulfate- or Sulfur-Reducing Bacteria

Genus *Desulfuromonas*
Genus *Desulfovibrio*
Genus *Desulfomonas*
Genus *Desulfococcus*
Genus *Desulfobacter*
Genus *Desulfobulbus*
Genus *Desulfosarcina*

Section 8

Anaerobic Gram-Negative Cocci

Family I *Veillonellaceae*
Genus I *Veillonella*
Genus II *Acidaminococcus*
Genus III *Megasphaera*

Section 9

The Rickettsias and Chlamydias

Order I *Rickettsiales*
Family I *Rickettsiaceae*
Tribe I *Rickettsieae*
Genus I *Rickettsia*
Genus II *Rochalimaea*
Genus III *Coxiella*
Tribe II *Ehrlichieae*
Genus IV *Ehrlichia*
Genus V *Cowdria*
Genus VI *Neorickettsia*
Tribe III *Wolbachieae*
Genus VII *Wolbachia*
Genus VIII *Rickettsiella*
Family II *Bartonellaceae*
Genus I *Bartonella*
Genus II *Grahamella*
Family III *Anaplasmataceae*
Genus I *Anaplasma*
Genus II *Aegyptianella*
Genus III *Haemobartonella*
Genus IV *Eperythrozoon*
Order II *Chlamydiales*
Family I *Chlamydiaceae*
Genus I *Chlamydia*

Section 10

The Mycoplasmas

Division *Tenericutes*

Class I *Mollicutes*

Order I *Mycoplasmatales*
Family I *Mycoplasmataceae*
Genus I *Mycoplasma*
Genus II *Ureaplasma*
Family II *Acholeplasmataceae*
Genus I *Acholeplasma*
Family III *Spiroplasmataceae*
Genus I *Spiroplasma*
Other Genera
Genus *Anaeroplasma*
Genus *Thermoplasma*
Mycoplasma-like Organisms of Plants and Invertebrates

Section 11

Endosymbionts

A. Endosymbionts of Protozoa
Endosymbionts of ciliates
Endosymbionts of flagellates
Endosymbionts of amoebas
Taxa of endosymbionts
Genus I *Holospora*
Genus II *Caedibacter*
Genus III *Pseudocaedibacter*
Genus IV *Lyticum*
Genus V *Tectibacter*
B. Endosymbionts of Insects
Bloodsucking insects
Plant sap-sucking insects
Cellulose and stored grain feeders
Insects feeding on complex diets
Taxon of endosymbionts:
Genus *Blattabacterium*
C. Endosymbionts of Fungi and Invertebrates Other Than Arthropods
Fungi
Sponges
Coelenterates
Helminthes
Annelids
Marine worms and mollusks

Volume II

Section 12

Gram-Positive Cocci

Family I *Micrococcaceae*
Genus I *Micrococcus*
Genus II *Stomatococcus*
Genus III *Planococcus*
Genus IV *Staphylococcus*
Family II *Deinococcaceae*
Genus I *Deinococcus*
Other Genera
Genus *Streptococcus*
Pyogenic Hemolytic Streptococci
Oral Streptococci
Enterococci
Lactic Acid Streptococci
Anaerobic Streptococci
Other Streptococci
Genus *Leuconostoc*
Genus *Pediococcus*
Genus *Aerococcus*
Genus *Gemella*
Genus *Peptococcus*
Genus *Peptostreptococcus*
Genus *Ruminococcus*
Genus *Coprococcus*
Genus *Sarcina*

Section 13

Endospore-Forming Gram-Positive Rods and Cocci

Genus *Bacillus*
Genus *Sporolactobacillus*
Genus *Clostridium*
Genus *Desulfotomaculum*
Genus *Sporosarcina*
Genus *Oscillospira*

Section 14

Regular, Nonsporing, Gram-Positive Rods

Genus *Lactobacillus*
Genus *Listeria*
Genus *Erysipelothrix*
Genus *Brochothrix*
Genus *Renibacterium*
Genus *Kurthia*
Genus *Caryophanon*

Section 15

Irregular, Nonsporing, Gram-Positive Rods

Genus *Corynebacterium*
Plant Pathogenic Species of *Corynebacterium*
Genus *Gardnerella*
Genus *Arcanobacterium*
Genus *Arthrobacter*
Genus *Brevibacterium*
Genus *Curtobacterium*
Genus *Caseobacter*
Genus *Microbacterium*

Genus *Aureobacterium*
Genus *Cellulomonas*
Genus *Agromyces*
Genus *Arachnia*
Genus *Rothia*
Genus *Propionibacterium*
Genus *Eubacterium*
Genus *Acetobacterium*
Genus *Lachnospira*
Genus *Butyrivibrio*
Genus *Thermoanaerobacter*
Genus *Actinomyces*
Genus *Bifidobacterium*

Section 16

The Mycobacteria

Family *Mycobacteriaceae*
Genus *Mycobacterium*

Section 17

Nocardioforms

Genus *Nocardia*
Genus *Rhodococcus*
Genus *Nocardioides*
Genus *Pseudonocardia*
Genus *Oerskovia*
Genus *Saccharopolyspora*
Genus *Micropolyspora*
Genus *Promicromonospora*
Genus *Intrasporangium*

Volume III

Section 18

Anoxygenic Phototrophic Bacteria

I. Purple Bacteria

Family I *Chromatiaceae*
Genus I *Chromatium*
Genus II *Thiocystis*
Genus III *Thiospirillum*
Genus IV *Thiocapsa*
Genus V *Lamprobacter*
Genus VI *Lamprocystis*
Genus VII *Thiodictyon*
Genus VIII *Amoebobacter*
Genus IX *Thiopedia*
Family II *Ectothiorhodospiraceae*
Genus *Ectothiorhodospira*
Purple Nonsulfur Bacteria
Genus *Rhodospirillum*
Genus *Rhodopila*
Genus *Rhodobacter*
Genus *Rhodopseudomonas*
Genus *Rhodomicrobium*
Genus *Rhodocyclus*

II. Green Bacteria

Green Sulfur Bacteria
Genus *Chlorobium*
Genus *Prosthecochloris*
Genus *Pelodictyon*
Genus *Ancalochloris*
Genus *Chloroherpeton*
Multicellular, Filamentous, Green Bacteria
Genus *Chloroflexus*
Genus *Heliothrix*
Genus "*Oscillochloris*"
Genus *Chloronema*

III. Genera Incertae Sedis

Genus *Heliobacterium*
Genus *Erythrobacter*

Section 19

Oxygenic Photosynthetic Bacteria

Group I Cyanobacteria

Subsection I Order *Chroococcales*
1. Genus I *Chamaesiphon*
2. Genus II *Gloeobacter*
3. *Synechococcus*-group
4. Genus III *Gloeothece*
5. *Cyanothece*-group
6. *Gloeocapsa*-group
7. *Synechocystis*-group

Subsection II Order *Pleurocapsales*
1. Genus I *Dermocarpa*
2. Genus II *Xenococcus*
3. Genus III *Dermocarpella*
4. Genus IV *Mxyosarcina*
5. Genus V *Chroococcidiopsis*
6. *Pleurocapsa*-group

Subsection III Order *Oscillatoriales*
Genus I *Spirulina*
Genus II *Arthrospira*
Genus III *Oscillatoria*
Genus IV *Lyngbya*
Genus V *Pseudanabaena*
Genus VI *Starria*
Genus VII *Crinalium*
Genus VIII *Microcoleus*
Subsection IV Order *Nostocales*
Family I *Nostocaceae*
Genus I *Anabaena*
Genus II *Aphanizomenon*
Genus III *Nodularia*
Genus IV *Cylindrospermum*
Genus V *Nostoc*
Family II *Scytonemataceae*
Genus I *Scytonema*
Family III *Rivulariaceae*
Genus I *Calothrix*
Subsection V Order *Stigonematales*
Genus I *Chlorogloeopsis*
Genus II *Fischerella*
Genus III *Stigonema*
Genus IV *Geitleria*

Group II Order Prochlorales

Family I *Prochloraceae*
Genus *Prochloron*
Other taxa
Genus "*Prochlorothrix*"

Section 20

Aerobic Chemolithotrophic Bacteria and Associated Organisms

A. Nitrifying Bacteria
Family *Nitrobacteraceae*
Nitrite-oxidizing bacteria
Genus I *Nitrobacter*
Genus II *Nitrospina*
Genus III *Nitrococcus*
Genus IV *Nitrospira*
Ammonia-oxidizing bacteria
Genus V *Nitrosomonas*
Genus VI *Nitrosococcus*
Genus VII *Nitrosospira*
Genus VIII *Nitrosolobus*
Genus IX "*Nitrosovibrio*"
B. Colorless Sulfur Bacteria
Genus *Thiobacterium*
Genus *Macromonas*
Genus *Thiospira*
Genus *Thiovulum*
Genus *Thiobacillus*
Genus *Thiomicrospira*
Genus *Thiosphaera*
Genus *Acidiphilium*
Genus *Thermothrix*
C. Obligately Chemolithotrophic Hydrogen Bacteria
Genus *Hydrogenobacter*
D. Iron- and Manganese-Oxidizing and/or Depositing Bacteria
Family "*Siderocapsaceae*"
Genus I "*Siderocapsa*"
Genus II "*Naumanniella*"
Genus III "*Siderococcus*"
Genus IV "*Ochrobium*"
E. Magnetotactic Bacteria
Genus *Aquaspirillum (A. magnetotacticum)*
Genus "*Bilophococcus*"

Section 21

Budding and/or Appendaged Bacteria

I. Prosthecate Bacteria

A. Budding bacteria
1. Buds produced at tip of prostheca
Genus *Hyphomicrobium*
Genus *Hyphomonas*
Genus *Pedomicrobium*
2. Buds produced on cell surface
Genus *Ancalomicrobium*
Genus *Prosthecomicrobium*
Genus *Labrys*
Genus *Stella*

B. Bacteria that divide by binary transverse fission
Genus *Caulobacter*
Genus *Asticcacaulis*
Genus *Prosthecobacter*

II. Nonprosthecate Bacteria

A. Budding bacteria
1. Lack peptidoglycan
Genus *Planctomyces*
Genus "*Isosphaera*"
2. Contain peptidoglycan
Genus *Ensifer*
Genus *Blastobacter*
Genus *Angulomicrobium*
Genus *Gemmiger*

B. Nonbudding, stalked bacteria
Genus *Gallionella*
Genus *Nevskia*

C. Other bacteria
1. Nonspinate bacteria
Genus *Seliberia*
Genus "*Metallogenium*"
Genus "*Thiodendron*"
2. Spinate bacteria

Section 22

Sheathed Bacteria

Genus *Sphaerotilus*
Genus *Leptothrix*
Genus *Haliscomenobacter*
Genus "*Lieskeella*"
Genus "*Phragmidiothrix*"
Genus *Crenothrix*
Genus "*Clonothrix*"

Section 23

Nonphotosynthetic, Nonfruiting Gliding Bacteria

Order I *Cytophagales*
Family I *Cytophagaceae*
Genus I *Cytophaga*
Genus II *Capnocytophaga*
Genus III *Flexithrix*
Genus IV *Sporocytophaga*
Other genera
Genus *Flexibacter*
Genus *Microscilla*
Genus *Chitinophaga*
Genus *Saprospira*
Order II *Lysobacterales*
Family I *Lysobacteraceae*
Genus I *Lysobacter*
Order III *Beggiatoales*
Family I *Beggiatoaceae*
Genus I *Beggiatoa*
Genus II *Thiothrix*
Genus III *Thioploca*
Genus IV "*Thiospirillopsis*"
Other families and genera
Family *Simonsiellaceae*
Genus I *Simonsiella*
Genus II *Alysiella*
Family "*Pelonemataceae*"
Genus I "*Pelonema*"
Genus II "*Achroonema*"
Genus III "*Peloploca*"
Genus IV "*Desmanthos*"
Other genera
Genus *Toxothrix*
Genus *Leucothrix*
Genus *Vitreoscilla*
Genus *Desulfonema*
Genus *Achromatium*
Genus *Agitococcus*
Genus *Herpetosiphon*

Section 24

Fruiting Gliding Bacteria: The Myxobacteria

Order *Myxococcales*
Family I *Myxococcaceae*
Genus *Myxococcus*
Family II *Archangiaceae*
Genus *Archangium*
Family III *Cystobacteraceae*
Genus I *Cystobacter*
Genus II *Melittangium*
Genus III *Stigmatella*
Family IV *Polyangiaceae*
Genus I *Polyangium*
Genus II *Nannocystis*
Genus III *Chondromyces*

Section 25

Archaeobacteria

Group I Methanogenic Archaeobacteria

Order I *Methanobacteriales*
Family I *Methanobacteriaceae*
Genus I *Methanobacterium*
Genus II *Methanobrevibacter*
Family II *Methanothermaceae*
Genus *Methanothermus*
Order II *Methanococcales*
Family *Methanococcaceae*
Genus *Methanococcus*
Order III *Methanomicrobiales*
Family I *Methanomicrobiaceae*
Genus I *Methanomicrobium*
Genus II *Methanospirillum*
Genus III *Methanogenium*
Family II *Methanosarcinaceae*
Genus I *Methanosarcina*
Genus II *Methanolobus*
Genus III *Methanothrix*
Genus IV *Methanococcoides*
Other Taxa
Family *Methanoplanaceae*
Genus *Methanoplanus*
Other Genus *Methanosphaera*

Group II Archaeobacterial Sulfate Reducers

Order "*Archaeoglobales*"
Family "*Archaeoglobaceae*"
Genus *Archaeoglobus*

Group III Extremely Halophilic Archaeobacteria

Order *Halobacteriales*
Family *Halobacteriaceae*
Genus I *Halobacterium*
Genus II *Haloarcula*
Genus III *Haloferax*
Genus IV *Halococcus*
Genus V *Natronobacterium*
Genus VI *Natronococcus*

Group IV Cell Wall-less Archaeobacteria

Genus *Thermoplasma*

Group V Extremely Thermophilic S^0-Metabolizers

Order I *Thermococcales*
Family *Thermococcaceae*
Genus I *Thermococcus*
Genus II *Pyrococcus*
Order II *Thermoproteales*
Family I *Thermoproteaceae*
Genus I *Thermoproteus*
Genus II *Thermofilum*
Family II *Desulfurococcaceae*
Genus *Desulfurococcus*
Other bacteria
Genus *Staphylothermus*
Genus *Pyrodictium*
Order III *Sulfolobales*
Family *Sulfolobaceae*
Genus I *Sulfolobus*
Genus II *Acidianus*

Volume IV

Section 26

Nocardioform Actinomycetes

Genus *Nocardia*
Genus *Rhodococcus*
Genus *Nocardioides*
Genus *Pseudonocardia*
Genus *Oerskovia*
Genus *Saccharopolyspora*
Genus *Faenia*
Genus *Promicromonospora*
Genus *Intrasporangium*
Genus *Actinopolyspora*
Genus *Saccharomonospora*

Section 27

Actinomycetes with Multilocular Sporangia

Genus *Geodermatophilus*
Genus *Dermatophilus*
Genus *Frankia*

Section 28

Actinoplanetes

Genus *Actinoplanes*
Genus *Ampullariella*
Genus *Pilimelia*

Genus *Dactylosporangium*
Genus *Micromonospora*

Section 29

Streptomycetes and Related Genera

Genus *Streptomyces*
Genus *Streptoverticillium*
Genus *Kineosporia*
Genus *Sporichthya*

Section 30

Maduromycetes

Genus *Actinomadura*
Genus *Microbispora*
Genus *Microtetraspora*
Genus *Planobispora*
Genus *Planomonospora*
Genus *Spirillospora*
Genus *Streptosporangium*

Section 31

Thermomonospora and Related Genera

Genus *Thermomonospora*
Genus *Actinosynnema*
Genus *Nocardiopsis*
Genus *Streptoalloteichus*

Section 32

Thermoactinomycetes

Genus *Thermoactinomyces*

Section 33

Other Genera

Genus *Glycomyces*
Genus *Kibdelosporangium*
Genus *Kitasatosporia*
Genus *Saccharothrix*
Genus *Pasteuria*

APPENDIX IV

Classification of Procaryotes According to the Second Edition of *Bergey's Manual of Systematic Bacteriology*

This appendix summarizes the classification of procaryotes that is being employed by the second edition of *Bergey's Manual.* The second edition will be published in five volumes over a period of several years beginning in 2001. Because of advances in taxonomy over the next few years, some of the details in this classification may well change. The overall organization and major taxa should remain constant. The quotation marks around some names indicate that they are not formally approved taxonomic names.

Domain *Archaea*

Phylum AI. *Crenarchaeota*

Class I. *Thermoprotei*

Order I. *Thermoproteales*
- Family I. *Thermoproteaceae*
  - Genus I. *Thermoproteus*
  - Genus II. *Caldivirga*
  - Genus III. *Pyrobaculum*
  - Genus IV. *Thermocladium*
- Family II. *Thermofilaceae*
  - Genus I. *Thermofilum*

Order II. *Desulfurococcales*
- Family I. *Desulfurococcaceae*
  - Genus I. *Desulfurococcus*
  - Genus II. *Aeropyrum*
  - Genus III. *Ignicoccus*
  - Genus IV. *Staphylothermus*
  - Genus V. *Stetteria*
  - Genus VI. *Sulfophobococcus*
  - Genus VII. *Thermodiscus*
  - Genus VIII. *Thermosphaera*
- Family II. *Pyrodictiaceae*
  - Genus I. *Pyrodictium*
  - Genus II. *Hyperthermus*
  - Genus III. *Pyrolobus*

Order III. *Sulfolobales*
- Family I. *Sulfolobaceae*
  - Genus I. *Sulfolobus*
  - Genus II. *Acidianus*
  - Genus III. *Metallosphaera*
  - Genus IV. *Stygiolobus*
  - Genus V. *Sulfurisphaera*
  - Genus VI. *Sulfurococcus*

Phylum AII. *Euryarchaeota*

Class I. *Methanobacteria*

Order I. *Methanobacteriales*
- Family I. *Methanobacteriaceae*
  - Genus I. *Methanobacterium*
  - Genus II. *Methanobrevibacter*
  - Genus III. *Methanosphaera*
  - Genus IV. *Methanothermobacter*
- Family II. *Methanothermaceae*
  - Genus I. *Methanothermus*

Class II. *Methanococci*

Order I. *Methanococcales*
- Family I. *Methanococcaceae*
  - Genus I. *Methanococcus*
  - Genus II. *Methanothermococcus*
- Family II. *Methanocaldococcaceae*
  - Genus I. *Methanocaldococcus*
  - Genus II. *Methanotorris*

Order II. *Methanomicrobiales*
- Family I. *Methanomicrobiaceae*
  - Genus I. *Methanomicrobium*
  - Genus II. *Methanoculleus*
  - Genus III. *Methanofollis*
  - Genus IV. *Methanogenium*
  - Genus V. *Methanolacinia*
  - Genus VI. *Methanoplanus*
- Family II. *Methanocorpusculaceae*
  - Genus I. *Methanocorpusculum*
- Family III. *Methanospirillaceae*
  - Genus I. *Methanospirillum*
- Genera incertae sedis
  - Genus I. *Methanocalculus*

Order III. *Methanosarcinales*
- Family I. *Methanosarcinaceae*
  - Genus I. *Methanosarcina*
  - Genus II. *Methanococcoides*
  - Genus III. *Methanohalobium*
  - Genus IV. *Methanohalophilus*
  - Genus V. *Methanolobus*
  - Genus VI. *Methanosalsum*
- Family II. *Methanosaetaceae*
  - Genus I. *Methanosaeta*

Class III. *Halobacteria*

Order I. *Halobacteriales*
- Family I. *Halobacteriaceae*
  - Genus I. *Halobacterium*
  - Genus II. *Haloarcula*
  - Genus III. *Halobaculum*
  - Genus IV. *Halococcus*
  - Genus V. *Haloferax*
  - Genus VI. *Halogeometricum*
  - Genus VII. *Halorhabdus*
  - Genus VIII. *Halorubrum*
  - Genus IX. *Haloterrigena*
  - Genus X. *Natrialba*
  - Genus XI. *Natrinema*
  - Genus XII. *Natronobacterium*
  - Genus XIII. *Natronococcus*
  - Genus XIV. *Natronomonas*
  - Genus XV. *Natronorubrum*

Class IV. *Thermoplasmata*

Order I. *Thermoplasmatales*
- Family I. *Thermoplasmataceae*
  - Genus I. *Thermoplasma*
- Family II. *Picrophilaceae*
  - Genus I. *Picrophilus*

Class V. *Thermococci*

Order I. *Thermococcales*
- Family I. *Thermococcaceae*
  - Genus I. *Thermococcus*
  - Genus II. *Pyrococcus*

Class VI. *Archaeoglobi*

Order I. *Archaeoglobales*
- Family I. *Archaeoglobaceae*
  - Genus I. *Archaeoglobus*
  - Genus II. *Ferroglobus*

Class VII. *Methanopyri*

Order I. *Methanopyrales*
- Family I. *Methanopyraceae*
  - Genus I. *Methanopyrus*

Domain *Bacteria*

Phylum BI. *Aquificae*

Class I. *Aquificae*

Order I. *Aquificales*
- Family I. *Aquificaceae*
  - Genus I. *Aquifex*
  - Genus II. *Calderobacterium*
  - Genus III. *Hydrogenobacter*
  - Genus IV. *Thermocrinis*
- Genera incertae sedis
  - Genus I. *Desulfurobacterium*

Phylum BII. *Thermotogae*

Class I. *Thermotogae*

Order I. *Thermotogales*

Family I. *Thermotogaceae*
Genus I. *Thermotoga*
Genus II. *Fervidobacterium*
Genus III. *Geotoga*
Genus IV. *Petrotoga*
Genus V. *Thermosipho*

Phylum BIII. *Thermodesulfobacteria*

Class I. *Thermodesulfobacteria*

Order I. *Thermodesulfobacteriales*
Family I. *Thermodesulfobacteriaceae*
Genus I. *Thermodesulfobacterium*

Phylum BIV. "Deinococcus-Thermus"

Class I. *Deinococci*

Order I. *Deinococcales*
Family I. *Deinococcaceae*
Genus I. *Deinococcus*
Order II. *Thermales*
Family I. *Thermaceae*
Genus I. *Thermus*
Genus II. *Meiothermus*

Phylum BV. *Chrysiogenetes*

Class I. *Chrysiogenetes*

Order I. *Chrysiogenales*
Family I. *Chrysiogenaceae*
Genus I. *Chrysiogenes*

Phylum BVI. *Chloroflexi*

Class I. "Chloroflexi"

Order I. "Chloroflexales"
Family I. "Chloroflexaceae"
Genus I. *Chloroflexus*
Genus II. *Chloronema*
Genus III. *Heliothrix*
Genus IV. *Oscillochloris*
Order II. "Herpetosiphonales"
Family I. "Herpetosiphonaceae"
Genus I. *Herpetosiphon*

Phylum BVII. *Thermomicrobia*

Class I. *Thermomicrobia*

Order I. *Thermomicrobiales*
Family I. *Thermomicrobiaceae*
Genus I. *Thermomicrobium*

Phylum BVIII. *Nitrospira*

Class I. "Nitrospira"

Order I. "Nitrospirales"
Family I. "Nitrospiraceae"
Genus I. *Nitrospira*
Genus II. *Leptospirillum*
Genus III. *Magnetobacterium*
Genus IV. *Thermodesulfovibrio*

Phylum BIX. *Deferribacteres*

Class I. *Deferribacteres*

Order I. *Deferribacterales*
Family I. *Deferribacteraceae*
Genus I. *Deferribacter*
Genus II. "Flexistipes"
Genus III. *Geovibrio*
Genera incertae sedis
Genus I. *Synergistes*

Phylum BX. *Cyanobacteria*

Class I. "Cyanobacteria"

Subsection I.
Genus I. *Chamaesiphon*
Genus II. *Chroococcus*
Genus III. *Cyanobacterium*
Form genus IV. *Cyanobium*
Genus V. *Cyanothece*
Genus VI. *Dactylococcopsis*
Genus VII. *Gloeobacter*
Genus VIII. *Gloeocapsa*
Genus IX. *Gloeothece*
Genus X. *Microcystis*
Genus XI. *Prochlorococcus*
Genus XII. *Prochloron*
Form genus XIII. *Synechococcus*
Form genus XIV. *Synechocystis*
Subsection II.
Subgroup I.
Genus I. *Cyanocystis*
Genus II. *Dermocarpella*
Genus III. *Stanieria*
Genus IV. *Xenococcus*
Subgroup II.
Genus I. *Chroococcidiopsis*
Genus II. *Myxosarcina*
Genus III. *Pleurocapsa*
Subsection III.
Genus I. *Arthrospira*
Genus II. *Borzia*
Genus III. *Crinalium*
Genus IV. *Geitlerinema*
Genus V. *Leptolyngbya*
Genus VI. *Limnothrix*
Genus VII. *Lyngbya*
Genus VIII. *Microcoleus*
Genus IX. *Oscillatoria*
Genus X. *Planktothrix*
Genus XI. *Prochlorothrix*
Genus XII. *Pseudanabaena*
Genus XIII. *Spirulina*
Genus XIV. *Starria*
Genus XV. *Symploca*
Genus XVI. *Trichodesmium*
Genus XVII. *Tychonema*
Subsection IV.
Subgroup I.
Genus I. *Anabaena*
Genus II. *Anabaenopsis*
Genus III. *Aphanizomenon*
Genus IV. *Cyanospira*
Genus V. *Cylindrospermopsis*
Genus VI. *Cylindrospermum*
Genus VII. *Nodularia*
Genus VIII. *Nostoc*
Genus IX. *Scytonema*
Subgroup II.
Genus I. *Calothrix*
Genus II. *Rivularia*
Genus III. *Tolypothrix*
Subsection V.
Family I.
Genus I. *Chlorogloeopsis*
Genus II. *Fischerella*
Genus III. *Geitleria*
Genus IV. *Iyengariella*
Genus V. *Nostochopsis*
Genus VI. *Stigonema*

Phylum BXI. *Chlorobi*

Class I. "Chlorobia"

Order I. *Chlorobiales*
Family I. *Chlorobiaceae*
Genus I. *Chlorobium*
Genus II. *Ancalochloris*
Genus III. *Chloroherpeton*
Genus IV. *Pelodictyon*
Genus V. *Prosthecochloris*

Phylum BXII. *Proteobacteria*

Class I. "Alphaproteobacteria"

Order I. *Rhodospirillales*
Family I. *Rhodospirillaceae*
Genus I. *Rhodospirillum*
Genus II. *Azospirillum*
Genus III. *Magnetospirillum*
Genus IV. *Phaeospirillum*
Genus V. *Rhodocista*
Genus VI. *Rhodospira*
Genus VII. *Rhodothalassium*
Genus VIII. *Rhodovibrio*
Genus IX. *Roseospira*
Genus X. *Skermanella*
Family II. *Acetobacteraceae*
Genus I. *Acetobacter*
Genus II. *Acidiphilium*
Genus III. *Acidocella*
Genus IV. *Acidomonas*
Genus V. *Craurococcus*
Genus VI. *Gluconacetobacter*
Genus VII. *Gluconobacter*
Genus VIII. *Paracraurococcus*
Genus IX. *Rhodopila*
Genus X. *Roseococcus*
Genus XI. *Stella*
Genus XII. *Zavarzinia*
Order II. *Rickettsiales*
Family I. *Rickettsiaceae*
Genus I. *Rickettsia*
Genus II. *Orientia*
Genus III. *Wolbachia*
Family II. *Ehrlichiaceae*
Genus I. *Ehrlichia*
Genus II. *Aegyptianella*
Genus III. *Anaplasma*
Genus IV. *Cowdria*
Genus V. *Neorickettsia*

Family III. "Holosporaceae"
Genus I. *Holospora*
Genus II. *Caedibacter*
Genus III. *Lyticum*
Genus IV. *Polynucleobacter*
Genus V. *Pseudocaedibacter*
Genus VI. *Symbiotes*
Genus VII. *Tectibacter*
Order III. "Rhodobacterales"
Family I. "Rhodobacteraceae"
Genus I. *Rhodobacter*
Genus II. *Ahrensia*
Genus III. *Amaricoccus*
Genus IV. *Antarctobacter*
Genus V. *Gemmobacter*
Genus VI. *Hirschia*
Genus VII. *Hyphomonas*
Genus VIII. *Maricaulis*
Genus IX. *Octadecabacter*
Genus X. *Paracoccus*
Genus XI. *Rhodovulum*
Genus XII. *Roseivivax*
Genus XIII. *Roseobacter*
Genus XIV. *Roseovarius*
Genus XV. *Rubrimonas*
Genus XVI. *Ruegeria*
Genus XVII. *Sagittula*
Genus XVIII. *Staleya*
Genus XIX. *Stappia*
Genus XX. *Sulfitobacter*
Order IV. "Sphingomonadales"
Family I. "Sphingomonadaceae"
Genus I. *Sphingomonas*
Genus II. *Blastomonas*
Genus III. *Erythrobacter*
Genus IV. *Erythromicrobium*
Genus V. *Erythromonas*
Genus VI. *Porphyrobacter*
Genus VII. *Rhizomonas*
Genus VIII. *Sandaracinobacter*
Genus IX. *Zymomonas*
Order V. *Caulobacterales*
Family I. *Caulobacteraceae*
Genus I. *Caulobacter*
Genus II. *Asticcacaulis*
Genus III. *Brevundimonas*
Genus IV. *Phenylobacterium*
Order VI. "Rhizobiales"
Family I. *Rhizobiaceae*
Genus I. *Rhizobium*
Genus II. *Agrobacterium*
Genus III. *Carbophilus*
Genus IV. *Chelatobacter*
Genus V. *Ensifer*
Genus VI. *Sinorhizobium*
Family II. *Bartonellaceae*
Genus I. *Bartonella*
Family III. *Brucellaceae*
Genus I. *Brucella*
Genus II. *Mycoplana*
Genus III. *Ochrobactrum*
Family IV. "Phyllobacteriaceae"
Genus I. *Phyllobacterium*
Genus II. *Allorhizobium*
Genus III. *Aminobacter*
Genus IV. *Aquamicrobium*
Genus V. *Defluvibacter*
Genus VI. *Mesorhizobium*
Genus VII. *Pseudaminobacter*
Family V. "Methylocystaceae"
Genus I. *Methylocystis*
Genus II. *Methylopila*
Genus III. *Methylosinus*
Family VI. "Beijerinckiaceae"
Genus I. *Beijerinckia*
Genus II. *Chelatococcus*
Genus III. *Derxia*
Family VII. "Bradyrhizobiaceae"
Genus I. *Bradyrhizobium"*
Genus II. *Afipia*
Genus III. *Agromonas*
Genus IV. *Blastobacter*
Genus V. *Bosea*
Genus VI. *Nitrobacter*
Genus VII. *Oligotropha*
Genus VIII. *Rhodopseudomonas*
Family VIII. *Hyphomicrobiaceae*
Genus I. *Hyphomicrobium*
Genus II. *Ancalomicrobium*
Genus III. *Ancylobacter*
Genus IV. *Angulomicrobium*
Genus V. *Aquabacter*
Genus VI. *Azorhizobium*
Genus VII. *Blastochloris*
Genus VIII. *Devosia*
Genus IX. *Dichotomicrobium*
Genus X. *Filomicrobium*
Genus XI. *Gemmiger*
Genus XII. *Labrys*
Genus XIII. *Methylorhabdus*
Genus XIV. *Pedomicrobium*
Genus XV. *Prosthecomicrobium*
Genus XVI. *Rhodomicrobium*
Genus XVII. *Rhodoplanes*
Genus XVIII. *Seliberia*
Genus XIX. *Xanthobacter*
Family IX. "Methylobacteriaceae"
Genus I. *Methylobacterium*
Genus II. *Protomonas*
Genus III. *Roseomonas*
Family X. "Rhodobiaceae"
Genus I. *Rhodobium*

Class II. "Betaproteobacteria"

Order I. "Burkholderiales"
Family I. "Burkholderiaceae"
Genus I. *Burkholderia*
Genus II. *Cupriavidus*
Genus III. *Lautropia*
Genus IV. *Thermothrix*
Family II. "Ralstoniaceae"
Genus I. *Ralstonia*
Family III. "Oxalobacteraceae"
Genus I. *Oxalobacter*
Genus II. *Duganella*
Genus III. *Herbaspirillum*
Genus IV. *Janthinobacterium*
Genus V. *Telluria*
Family IV. *Alcaligenaceae*
Genus I. *Alcaligenes*
Genus II. *Achromobacter*
Genus III. *Bordetella*
Genus IV. *Pelistega*
Genus V. *Sutterella*
Genus VI. *Taylorella*
Family V. *Comamonadaceae*
Genus I. *Comamonas*
Genus II. *Acidovorax*
Genus III. *Aquabacterium*
Genus IV. *Brachymonas*
Genus V. *Delftia*
Genus VI. *Hydrogenophaga*
Genus VII. *Ideonella*
Genus VIII. *Leptothrix*
Genus IX. *Polaromonas*
Genus X. *Rhodoferax*
Genus XI. *Roseateles*
Genus XII. *Rubrivivax*
Genus XIII. *Sphaerotilus*
Genus XIV. *Thiomonas*
Genus XV. *Variovorax*
Order II. "Hydrogenophilales"
Family I. "Hydrogenophilaceae"
Genus I. *Hydrogenophilus*
Genus II. *Thiobacillus*
Order III. "Methylophilales"
Family I. "Methylophilaceae"
Genus I. *Methylophilus*
Genus II. *Methylobacillus*
Genus III. *Methylovorus*
Order IV. "Neisseriales"
Family I. *Neisseriaceae*
Genus I. *Neisseria*
Genus II. *Alysiella*
Genus III. *Aquaspirillum*
Genus IV. *Catenococcus*
Genus V. *Chromobacterium*
Genus VI. *Eikenella*
Genus VII. *Formivibrio*
Genus VIII. *Iodobacter*
Genus IX. *Kingella*
Genus X. *Microvirgula*
Genus XI. *Prolinoborus*
Genus XII. *Simonsiella*
Genus XIII. *Vitreoscilla*
Genus XIV. *Vogesella*
Order V. "Nitrosomonadales"
Family I. "Nitrosomonadaceae"
Genus I. *Nitrosomonas*
Genus II. *Nitrosospira*
Family II. *Spirillaceae*
Genus I. *Spirillum*

Family III. *Gallionellaceae*
Genus I. *Gallionella*
Order VI. "Rhodocyclales"
Family I. *Rhodocyclaceae*
Genus I. *Rhodocyclus*
Genus II. *Azoarcus*
Genus III. *Propionibacter*
Genus IV. *Propionivibrio*
Genus V. *Thauera*
Genus VI. *Zoogloea*

Class III. "Gammaproteobacteria"

Order I. "Chromatiales"
Family I. *Chromatiaceae*
Genus I. *Chromatium*
Genus II. *Allochromatium*
Genus III. *Amoebobacter*
Genus IV. *Halochromatium*
Genus V. *Isochromatium*
Genus VI. *Lamprobacter*
Genus VII. *Lamprocystis*
Genus VIII. *Marichromatium*
Genus IX. *Nitrosococcus*
Genus X. *Pfennigia*
Genus XI. *Rhabdochromatium*
Genus XII. *Thermochromatium*
Genus XIII. *Thiocapsa*
Genus XIV. *Thiococcus*
Genus XV. *Thiocystis*
Genus XVI. *Thiodictyon*
Genus XVII. *Thiohalocapsa*
Genus XVIII. *Thiolamprovum*
Genus XIX. *Thiopedia*
Genus XX. *Thiorhodococcus*
Genus XXI. *Thiorhodovibrio*
Genus XXII. *Thiospirillum*
Family II. *Ectothiorhodospiraceae*
Genus I. *Ectothiorhodospira*
Genus II. *Arhodomonas*
Genus III. *Halorhodospira*
Genus IV. *Nitrococcus*
Genus V. *Thiorhodospira*
Order II. "Xanthomonadales"
Family I. "Xanthomonadaceae"
Genus I. *Xanthomonas*
Genus II. *Frateuria*
Genus III. *Lysobacter*
Genus IV. *Nevskia*
Genus V. *Pseudoxanthomonas*
Genus VI. *Rhodanobacter*
Genus VII. *Stenotrophomonas*
Genus VIII. *Xylella*
Order III. "Cardiobacteriales"
Family I. *Cardiobacteriaceae*
Genus I. *Cardiobacterium*
Genus II. *Dichelobacter*
Genus III. *Suttonella*
Order IV. "Thiotrichales"
Family I. "Thiotrichaceae"
Genus I. *Thiothrix*
Genus II. *Achromatium*
Genus III. *Beggiatoa*
Genus IV. *Leucothrix*
Genus V. *Macromonas*
Genus VI. *Thiobacterium*
Genus VII. *Thiomargarita*
Genus VIII. *Thioploca*
Genus IX. *Thiospira*
Family II. "Piscirickettsiaceae"
Genus I. *Piscirickettsia*
Genus II. *Cycloclasticus*
Genus III. *Hydrogenovibrio*
Genus IV. *Methylophaga*
Genus V. *Thiomicrospira*
Family III. "Francisellaceae"
Genus I. *Francisella*
Order V. "Legionellales"
Family I. *Legionellaceae*
Genus I. *Legionella*
Family II. "Coxiellaceae"
Genus I. *Coxiella*
Genus II. *Rickettsiella*
Order VI. "Methylococcales"
Family I. *Methylococcaceae*
Genus I. *Methylococcus*
Genus II. *Methylobacter*
Genus III. *Methylocaldum*
Genus IV. *Methylomicrobium*
Genus V. *Methylomonas*
Genus VI. *Methylosphaera*
Order VII. "Oceanospirillales"
Family I. "Oceanospirillaceae"
Genus I. *Oceanospirillum*
Genus II. *Balneatrix*
Genus III. *Fundibacter*
Genus IV. *Marinomonas*
Genus V. *Marinospirillum*
Genus VI. *Neptunomonas*
Family II. *Halomonadaceae*
Genus I. *Halomonas*
Genus II. *Alcanivorax*
Genus III. *Carnimonas*
Genus IV. *Chromohalobacter*
Genus V. *Deleya*
Genus VI. *Zymobacter*
Order VIII. *Pseudomonadales*
Family I. *Pseudomonadaceae*
Genus I. *Pseudomonas*
Genus II. *Azomonas*
Genus III. *Azotobacter*
Genus IV. *Cellvibrio*
Genus V. *Chryseomonas*
Genus VI. *Flavimonas*
Genus VII. *Lampropedia*
Genus VIII. *Mesophilobacter*
Genus IX. *Morococcus*
Genus X. *Oligella*
Genus XI. *Rhizobacter*
Genus XII. *Rugamonas*
Genus XIII. *Serpens*
Genus XIV. *Thermoleophilum*
Genus XV. *Xylophilus*
Family II. *Moraxellaceae*
Genus I. *Moraxella*
Genus II. *Acinetobacter*
Genus III. *Psychrobacter*
Order IX. "Alteromonadales"
Family I. "Alteromonadaceae"
Genus I. *Alteromonas*
Genus II. *Colwellia*
Genus III. *Ferrimonas*
Genus IV. *Glaciecola*
Genus V. *Marinobacter*
Genus VI. *Marinobacterium*
Genus VII. *Microbulbifer*
Genus VIII. *Moritella*
Genus IX. *Pseudoalteromonas*
Genus X. *Shewanella*
Order X. "Vibrionales"
Family I. *Vibrionaceae*
Genus I. *Vibrio*
Genus II. *Allomonas*
Genus III. *Enhydrobacter*
Genus IV. *Listonella*
Genus V. *Photobacterium*
Genus VI. *Salinivibrio*
Order XI. "Aeromonadales"
Family I. *Aeromonadaceae*
Genus I. *Aeromonas*
Genus II. *Tolumonas*
Family II. *Succinivibrionaceae*
Genus I. *Succinivibrio*
Genus II. *Anaerobiospirillum*
Genus III. *Ruminobacter*
Genus IV. *Succinimonas*
Order XII. "Enterobacteriales"
Family I. *Enterobacteriaceae*
Genus I. *Enterobacter*
Genus II. *Alterococcus*
Genus III. *Arsenophonus*
Genus IV. *Brenneria*
Genus V. *Buchnera*
Genus VI. *Budvicia*
Genus VII. *Buttiauxella*
Genus VIII. *Calymmatobacterium*
Genus IX. *Cedecea*
Genus X. *Citrobacter*
Genus XI. *Edwardsiella*
Genus XII. *Erwinia*
Genus XIII. *Escherichia*
Genus XIV. *Ewingella*
Genus XV. *Hafnia*
Genus XVI. *Klebsiella*
Genus XVII. *Kluyvera*
Genus XVIII. *Leclercia*
Genus XIX. *Leminorella*
Genus XX. *Moellerella*
Genus XXI. *Morganella*
Genus XXII. *Obesumbacterium*
Genus XXIII. *Pantoea*
Genus XXIV. *Pectobacterium*
Genus XXV. *Photorhabdus*
Genus XXVI. *Plesiomonas*

Genus XXVII. *Pragia*
Genus XXVIII. *Proteus*
Genus XXIX. *Providencia*
Genus XXX. *Rahnella*
Genus XXXI. *Saccharobacter*
Genus XXXII. *Salmonella*
Genus XXXIII. *Serratia*
Genus XXXIV. *Shigella*
Genus XXXV. *Sodalis*
Genus XXXVI. *Tatumella*
Genus XXXVII. *Trabulsiella*
Genus XXXVIII. *Wigglesworthia*
Genus XXXIX. *Xenorhabdus*
Genus XL. *Yersinia*
Genus XLI. *Yokenella*
Order XIII. "Pasteurellales"
Family I. *Pasteurellaceae*
Genus I. *Pasteurella*
Genus II. *Actinobacillus*
Genus III. *Haemophilus*
Genus IV. *Lonepinella*
Genus V. *Mannheimia*
Genus VI. *Phocoenobacter*

Class IV. "Deltaproteobacteria"

Order I. "Desulfurellales"
Family I. "Desulfurellaceae"
Genus I. *Desulfurella*
Genus II. *Hippea*
Order II. "Desulfovibrionales"
Family I. "Desulfovibrionaceae"
Genus I. *Desulfovibrio*
Genus II. *Bilophila*
Genus III. *Lawsonia*
Family II. "Desulfomicrobiaceae"
Genus I. *Desulfomicrobium"*
Family III. "Desulfohalobiaceae"
Genus I. *Desulfohalobium*
Genus II. *Desulfomonas*
Genus III. *Desulfonatronovibrio*
Order III. "Desulfobacterales"
Family I. "Desulfobacteraceae"
Genus I. *Desulfobacter*
Genus II. *Desulfobacterium*
Genus III. "Desulfobacula"
Genus IV. "Desulfobotulus"
Genus V. *Desulfocella*
Genus VI. *Desulfococcus*
Genus VII. *Desulfofaba*
Genus VIII. *Desulfofrigus*
Genus IX. *Desulfonema*
Genus X. *Desulfosarcina*
Genus XI. *Desulfospira*
Genus XII. *Desulfotalea*
Family II. "Desulfobulbaceae"
Genus I. *Desulfobulbus*
Genus II. *Desulfocapsa*
Genus III. *Desulfofustis*
Genus IV. *Desulforhopalus*
Family III. "Desulfoarculaceae"
Genus I. "Desulfoarculus"
Genus II. *Nitrospina*
Genus III. *Desulfobacca*
Genus IV. *Desulfomonile*
Order IV. "Desulfuromonadales"
Family I. "Desulfuromonadaceae"
Genus I. *Desulfuromonas*
Genus II. *Desulfuromusa*
Family II. "Geobacteraceae"
Genus I. *Geobacter*
Family III. "Pelobacteraceae
Genus I. *Pelobacter*
Genus II. *Malonomonas*
Order V. "Syntrophobacterales"
Family I. "Syntrophobacteraceae"
Genus I. *Syntrophobacter*
Genus II. *Desulfacinum*
Genus III. *Desulforhabdus*
Genus IV. *Thermodesulforhabdus*
Family II. "Syntrophaceae"
Genus I. *Syntrophus*
Genus II. *Smithella*
Order VI. "Bdellovibrionales"
Family I. "Bdellovibrionaceae"
Genus I. *Bdellovibrio*
Genus II. *Micavibrio*
Genus III. *Vampirovibrio*
Order VII. *Myxococcales*
Family I. *Myxococcaceae*
Genus I. *Myxococcus*
Genus II. *Angiococcus*
Family II. *Archangiaceae*
Genus I. *Archangium*
Family III. *Cystobacteraceae*
Genus I. *Cystobacter*
Genus II. *Melittangium*
Genus III. *Stigmatella*
Family IV. *Polyangiaceae*
Genus I. *Polyangium*
Genus II. *Chondromyces*
Genus III. *Nannocystis*

Class V. "Epsilonproteobacteria"

Order I. "Campylobacterales"
Family I. *Campylobacteraceae*
Genus I. *Campylobacter*
Genus II. *Arcobacter*
Genus III. *Sulfurospirillum*
Genus IV. *Thiovulum*
Family II. "Helicobacteraceae"
Genus I. *Helicobacter*
Genus II. *Wolinella*

Phylum BXIII. *Firmicutes*

Class I. "Clostridia"

Order I. *Clostridiales*
Family I. *Clostridiaceae*
Genus I. *Clostridium*
Genus II. *Acetivibrio*
Genus III. *Acidaminobacter*
Genus IV. *Anaerobacter*
Genus V. *Caloramator*
Genus VI. *Natronincola*
Genus VII. *Oxobacter*
Genus VIII. *Sarcina*
Genus IX. *Sporobacter*
Genus X. *Thermobrachium*
Genus XI. *Tindallia*
Family II. "Lachnospiraceae"
Genus I. *Lachnospira*
Genus II. *Acetitomaculum*
Genus III. *Anaerofilum*
Genus IV. *Butyrivibrio*
Genus V. *Catonella*
Genus VI. *Coprococcus*
Genus VII. *Johnsonella*
Genus VIII. *Pseudobutyrivibrio*
Genus IX. *Roseburia*
Genus X. *Ruminococcus*
Genux XI. *Sporobacterium*
Family III. "Peptostreptococcaceae"
Genus I. *Peptostreptococcus*
Genus II. *Filifactor*
Genus III. *Fusibacter*
Genus IV. *Helococcus*
Genus V. *Tissierella*
Family IV. "Eubacteriaceae"
Genus I. *Eubacterium*
Genus II. *Acetobacterium*
Genus III. *Pseudoramibacter*
Family V. *Peptococcaceae*
Genus I. *Peptococcus*
Genus II. *Anaeroarcus*
Genus III. *Anaerosinus*
Genus IV. *Anaerovibrio*
Genus V. *Carboxydothermus*
Genus VI. *Centipeda*
Genus VII. *Dehalobacter*
Genus VIII. *Dendrosporobacter*
Genus IX. *Desulfitobacterium*
Genus X. *Desulfonispora*
Genus XI. *Desulfosporosinus*
Genus XII.*Desulfotomaculum*
Genus XIII. *Mitsuokella*
Genus XIV. *Propionispira*
Genus XV. *Succinispira*
Genus XVI. *Syntrophobotulus*
Genus XVII. *Thermoterrabacterium*
Family VI. "Heliobacteriaceae"
Genus I. *Heliobacterium*
Genus II. *Heliobacillus*
Genus III. *Heliophilum*
Family VII. "Acidaminococcaceae"
Genus I. *Acidaminococcus*
Genus II. *Acetonema*
Genus III. *Anaeromusa*
Genus IV. *Dialister*
Genus V. *Megasphaera*
Genus VI. *Pectinatus*
Genus VII. *Phascolarctobacterium*
Genus VIII. *Quinella*
Genus IX. *Schwartzia*
Genus X. *Selenomonas*

Genus XI. *Sporomusa*
Genus XII. *Succiniclasticum*
Genus XIII. *Veillonella*
Genus XIV. *Zymophilus*
Family VIII. *Syntrophomonadaceae*
Genus I. *Syntrophomonas*
Genus II. *Acetogenium*
Genus III. *Aminobacterium*
Genus IV. *Aminomonas*
Genus V. *Anaerobaculum*
Genus VI. *Anaerobranca*
Genus VII. *Caldicellulosiruptor*
Genus VIII. *Dethiosulfovibrio*
Genus IX. *Syntrophospora*
Genus X. *Thermaerobacter*
Genus XI. *Thermanaerovibrio*
Genus XII. *Thermohydrogenium*
Genus XIII. *Thermosyntropha*
Order II. "Thermoanaerobacteriales"
Family I. "Thermoanaerobacteriaceae"
Genus I. *Thermoanaerobacterium*
Genus II. *Ammonifex*
Genus III. *Coprothermobacter*
Genus IV. *Moorella*
Genus V. *Sporotomaculum*
Genus VI. *Thermoanaerobacter*
Genus VII. *Thermoanaerobium*
Order III. *Haloanaerobiales*
Family I. *Haloanaerobiaceae*
Genus I. *Haloanaerobium*
Genus II. *Halocella*
Genus III. *Halothermothrix*
Genus IV. *Natroniella*
Family II. *Halobacteroidaceae*
Genus I. *Halobacteroides*
Genus II. *Acetohalobium*
Genus III. *Haloanaerobacter*
Genus IV. *Orenia*
Genus V. *Sporohalobacter*

Class II. *Mollicutes*

Order I. *Mycoplasmatales*
Family I. *Mycoplasmataceae*
Genus I. *Mycoplasma*
Genus II. *Eperythrozoon*
Genus III. *Haemobartonella*
Genus IV. *Ureaplasma*
Order II. *Entomoplasmatales*
Family I. *Entomoplasmataceae*
Genus I. *Entomoplasma*
Genus II. *Mesoplasma*
Family II. *Spiroplasmataceae*
Genus I. *Spiroplasma*
Order III. *Acholeplasmatales*
Family I. *Acholeplasmataceae*
Genus I. *Acholeplasma*
Order IV. *Anaeroplasmatales*
Family I. *Anaeroplasmataceae*
Genus I. *Anaeroplasma*
Genus II. *Asteroleplasma*
Order V. Incertae sedis
Family I. "Erysipelotrichaceae"
Genus I. *Erysipelothrix*
Genus II. *Holdemania*

Class III. "Bacilli"

Order I. *Bacillales*
Family I. *Bacillaceae*
Genus I. *Bacillus*
Genus II. *Amphibacillus*
Genus III. *Exiguobacterium*
Genus IV. *Gracilibacillus*
Genus V. *Halobacillus*
Genus VI. *Saccharococcus*
Genus VII. *Salibacillus*
Genus VIII. *Virgibacillus*
Family II. *Planococcaceae*
Genus I. *Planococcus*
Genus II. *Filibacter*
Genus III. *Kurthia*
Genus IV. *Sporosarcina*
Family III. *Caryophanaceae*
Genus I. *Caryophanon*
Family IV. "Listeriaceae"
Genus I. *Listeria*
Genus II. *Brochothrix*
Family V. "Staphylococcaceae"
Genus I. *Staphylococcus*
Genus II. *Gemella*
Genus III. *Macrococcus*
Genus IV. *Salinicoccus*
Family VI. "Sporolactobacillaceae"
Genus I. *Sporolactobacillus*
Genus II. *Marinococcus*
Family VII. "Paenibacillaceae"
Genus I. *Paenibacillus*
Genus II. *Ammoniphilus*
Genus III. *Aneurinibacillus*
Genus IV. *Brevibacillus*
Genus V. *Oxalophagus*
Genus VI. *Thermobacillus*
Family VIII. "Alicyclobacillaceae"
Genus I. *Alicyclobacillus*
Genus II. *Pasteuria*
Genus III. *Sulfobacillus*
Family IX. "Thermoactinomycetaceae"
Genus I. *Thermoactinomyces*
Order II. "Lactobacillales"
Family I. *Lactobacillaceae*
Genus I. *Lactobacillus*
Genus II. *Pediococcus*
Family II. "Aerococcaceae"
Genus I. *Aerococcus*
Genus II. *Abiotrophia*
Genus III. *Dolosicoccus*
Genus IV. *Eremococcus*
Genus V. *Facklamia*
Genus VI. *Globicatella*
Genus VII. *Ignavigranum*
Family III. "Carnobacteriaceae"
Genus I. *Carnobacterium*
Genus II. *Agitococcus*
Genus III. *Alloiococcus*
Genus IV. *Desemzia*
Genus V. *Dolosigranulum*
Genus VI. *Lactosphaera*
Genus VII. *Trichococcus*
Family IV. "Enterococcaceae"
Genus I. *Enterococcus*
Genus II. *Melissococcus*
Genus III. *Tetragenococcus*
Genus IV. *Vagococcus*
Family V. "Leuconostocaceae"
Genus I. *Leuconostoc*
Genus II. *Oenococcus*
Genus III. *Weissella*
Family VI. *Streptococcaceae*
Genus I. *Streptococcus*
Genus II. *Lactococcus*
Family VII. Incertae sedis
Genus I. *Acetoanaerobium*
Genus II. *Oscillospira*
Genus III. *Syntrophococcus*

Phylum BXIV. *Actinobacteria*

Class I. *Actinobacteria*

Subclass I. *Acidimicrobidae*
Order I. *Acidimicrobiales*
Suborder I. "Acidimicrobineae"
Family I. *Acidimicrobiaceae*
Genus I. *Acidimicrobium*
Subclass II. *Rubrobacteridae*
Order I. *Rubrobacterales*
Suborder I. "Rubrobacterineae"
Family I. *Rubrobacteraceae*
Genus I. *Rubrobacter*
Subclass III. *Coriobacteridae*
Order I. *Coriobacteriales*
Suborder I. "Coriobacterineae"
Family I. *Coriobacteriaceae*
Genus I. *Coriobacterium*
Genus II. *Atopobium*
Genus III. *Collinsella*
Genus IV. *Crytobacterium*
Genus V. *Eggerthella*
Genus VI. *Slackia*
Subclass IV. *Sphaerobacteridae*
Order I. *Sphaerobacterales*
Suborder I. "Sphaerobacterineae"
Family I. *Sphaerobacteraceae*
Genus I. *Sphaerobacter*
Subclass V. *Actinobacteridae*
Order I. *Actinomycetales*
Suborder I. *Actinomycineae*
Family I. *Actinomycetaceae*
Genus I. *Actinomyces*
Genus II. *Actinobaculum*
Genus III. *Arcanobacterium*
Genus IV. *Mobiluncus*
Suborder VI. *Micrococcineae*
Family I. *Micrococcaceae*
Genus I. *Micrococcus*

Genus II. *Arthrobacter*
Genus III. *Bogoriella*
Genus IV. *Demetria*
Genus V. *Kocuria*
Genus VI. *Leucobacter*
Genus VII. *Nesterenkonia*
Genus VIII. *Renibacterium*
Genus IX. *Rothia*
Genus X. *Stomatococcus*
Genus XI. *Terracoccus*
Family II. *Brevibacteriaceae*
Genus I. *Brevibacterium*
Family III. *Cellulomonadaceae*
Genus I. *Cellulomonas*
Genus II. *Oerskovia*
Genus III. *Rarobacter*
Family IV. *Dermabacteraceae*
Genus I. *Dermabacter*
Genus II. *Brachybacterium*
Family V. *Dermatophilaceae*
Genus I. *Dermatophilus*
Genus II. *Dermacoccus*
Genus III. *Kytococcus*
Family VI. *Intrasporangiaceae*
Genus I. *Intrasporangium*
Genus II. *Janibacter*
Genus III. *Ornithinicoccus*
Genus IV. *Sanguibacter*
Genus V. *Terrabacter*
Family VII. *Jonesiaceae*
Genus I. *Jonesia*
Family VIII. *Microbacteriaceae*
Genus I. *Microbacterium*
Genus II. *Agrococcus*
Genus III. *Agromyces*
Genus IV. *Aureobacterium*
Genus V. *Clavibacter*
Genus VI. *Cryobacterium*
Genus VII. *Curtobacterium*
Genus VIII. *Frigoribacterium*
Genus IX. *Leifsonia*
Genus X. *Rathayibacter*
Family IX. "Beutenbergiaceae"
Genus I. *Beutenbergia*
Family X. *Promicromonosporaceae*
Genus I. *Promicromonospora*
Suborder VII. *Corynebacterineae*
Family I. *Corynebacteriaceae*
Genus I. *Corynebacterium*
Family II. *Dietziaceae*
Genus I. *Dietzia*
Family III. *Gordoniaceae*
Genus I. *Gordonia*
Genus II. *Skermania*
Family IV. *Mycobacteriaceae*
Genus I. *Mycobacterium*
Family V. *Nocardiaceae*
Genus I. *Nocardia*
Genus II. *Rhodococcus*
Family VI. *Tsukamurellaceae*
Genus I. *Tsukamurella*
Family VII. "Williamsiaceae"
Genus I. *Williamsia*
Suborder VIII. *Micromonosporineae*
Family I. *Micromonosporaceae*
Genus I. *Micromonospora*
Genus II. *Actinoplanes*
Genus III. *Catellatospora*
Genus IV. *Catenuloplanes*
Genus V. *Couchioplanes*
Genus VI. *Dactylosporangium*
Genus VII. *Pilimelia*
Genus VIII. *Spirilliplanes*
Genus IX. *Verrucosispora*
Suborder IX. *Propionibacterineae*
Family I. *Propionibacteriaceae*
Genus I. *Propionibacterium*
Genus II. *Luteococcus*
Genus III. *Microlunatus*
Genus IV. *Propioniferax*
Genus V. *Tessaracoccus*
Family II. *Nocardioidaceae*
Genus I. *Nocardioides*
Genus II. *Aeromicrobium*
Genus III. *Friedmanniella*
Genus IV. *Kribbella*
Genus V. *Micropruina*
Suborder X. *Pseudonocardineae*
Family I. *Pseudonocardiaceae*
Genus I. *Pseudonocardia*
Genus II. *Actinopolyspora*
Genus III. *Amycolatopsis*
Genus IV. *Kibdelosporangium*
Genus V. *Kutzneria*
Genus VI. *Prauserella*
Genus VII. *Saccharomonospora*
Genus VIII. *Saccharopolyspora*
Genus IX. *Streptoalloteichus*
Genus X. *Thermobispora*
Genus XI. *Thermocrispum*
Family II. *Actinosynnemataceae*
Genus I. *Actinosynnema*
Genus II. *Actinokineospora*
Genus III. *Lentzea*
Genus IV. *Saccharothrix*
Suborder XI. *Streptomycineae*
Family I. *Streptomycetaceae*
Genus I. *Streptomyces*
Genus II. *Kitasatospora*
Genus III. *Streptoverticillium*
Suborder XII. *Streptosporangineae*
Family I. *Streptosporangiaceae*
Genus I. *Streptosporangium*
Genus II. *Herbidospora*
Genus III. *Microbispora*
Genus IV. *Microtetraspora*
Genus V. *Nonomuraea*
Genus VI. *Planobispora*
Genus VII. *Planomonospora*
Genus VIII. *Planopolyspora*
Genus IX. *Planotetraspora*
Family II. *Nocardiopsaceae*
Genus I. *Nocardiopsis*
Genus II. *Thermobifida*
Family III. *Thermomonosporaceae*
Genus I. *Thermomonospora*
Genus II. *Actinomadura*
Genus III. *Spirillospora*
Suborder XIII. *Frankineae*
Family I. *Frankiaceae*
Genus I. *Frankia*
Family II. *Geodermatophilaceae*
Genus I. *Geodermatophilus*
Genus II. *Blastococcus*
Genus III. *Modestobacter*
Family III. *Microsphaeraceae*
Genus I. *Microsphaera*
Family IV. *Sporichthyaceae*
Genus I. *Sporichthya*
Family V. *Acidothermaceae*
Genus I. *Acidothermus*
Family VI. "Kineosporiaceae"
Genus I. *Kineosporia*
Genus II. *Cryptosporangium*
Genus III. *Kineococcus*
Suborder XIV. *Glycomycineae*
Family I. *Glycomycetaceae*
Genus I. *Glycomyces*
Order II. *Bifidobacteriales*
Family I. *Bifidobacteriaceae*
Genus I. *Bifidobacterium*
Genus II. *Falcivibrio*
Genus III. *Gardnerella*
Family II. Unknown Affiliation
Genus I. *Actinobispora*
Genus II. *Actinocorallia*
Genus III. *Excellospora*
Genus IV. *Pelczaria*
Genus V. *Turicella*

Phylum BXV. *Planctomycetes*

Class I. "Planctomycetacia"

Order I. *Planctomycetales*
Family I. *Planctomycetaceae*
Genus I. *Planctomyces*
Genus II. *Gemmata*
Genus III. *Isosphaera*
Genus IV. *Pirellula*

Phylum BXVI. *Chlamydiae*

Class I. "Chlamydiae"

Order I. *Chlamydiales*
Family I. *Chlamydiaceae*
Genus I. *Chlamydia*
Genus II. *Chlamydophila*
Family II. *Parachlamydiaceae*
Genus I. *Parachlamydia*
Family III. *Simkaniaceae*
Genus I. *Simkania*
Family IV. *Waddliaceae*
Genus I. *Waddlia*

Phylum BXVII. *Spirochaetes*

Class I. "Spirochaetes"

Order I. *Spirochaetales*
- Family I. *Spirochaetaceae*
  - Genus I. *Spirochaeta*
  - Genus II. *Borrelia*
  - Genus III. *Brevinema*
  - Genus IV. *Clevelandina*
  - Genus V. *Cristispira*
  - Genus VI. *Diplocalyx*
  - Genus VII. *Hollandina*
  - Genus VIII. *Pillotina*
  - Genus IX. *Treponema*
- Family II. "Serpulinaceae"
  - Genus I. *Brachyspira*
  - Genus II. *Serpulina*
- Family III. *Leptospiraceae*
  - Genus I. *Leptonema*
  - Genus II. *Leptospira*

Phylum BXVIII. *Fibrobacteres*

Class I. "Fibrobacteres"

Order I. "Fibrobacterales"
- Family I. "Fibrobacteraceae"
  - Genus I. *Fibrobacter*

Phylum BXIX. *Acidobacteria*

Class I. "Acidobacteria"

Order I. "Acidobacteriales"
- Family I. "Acidobacteriaceae"
  - Genus I. *Acidobacterium*
  - Genus II. *Geothrix*
  - Genus III. *Holophaga*

Phylum BXX. *Bacteroidetes*

Class I. "Bacteroides"

Order I. "Bacteroidales"
- Family I. *Bacteroidaceae*
  - Genus I. *Bacteroides*
  - Genus II. *Acetofilamentum*
  - Genus III. *Acetomicrobium*
  - Genus IV. *Acetothermus*
  - Genus V. *Anaerorhabdus*
  - Genus VI. *Megamonas*
- Family II. "Rikenellaceae"
  - Genus I. *Rikenella*
  - Genus II. *Marinilabilia*
- Family III. "Porphyromonadaceae"
  - Genus I. *Porphyromonas*
- Family IV. "Prevotellaceae"
  - Genus I. *Prevotella*

Class II. "Flavobacteria"

Order I. "Flavobacteriales"
- Family I. *Flavobacteriaceae*
  - Genus I. *Flavobacterium*
  - Genus II. *Bergeyella*
  - Genus III. *Capnocytophaga*
  - Genus IV. *Cellulophaga*
  - Genus V. *Chryseobacterium*
  - Genus VI. *Coenonia*
  - Genus VII. *Empedobacter*
  - Genus VIII. *Gelidibacter*
  - Genus IX. *Ornithobacterium*
  - Genus X. *Polaribacter*
  - Genus XI. *Psychroflexus*
  - Genus XII. *Psychroserpens*
  - Genus XIII. *Riemerella*
  - Genus XIV. *Weeksella*
- Family II. "Myroidaceae"
  - Genus I. *Myroides*
  - Genus II. *Psychromonas*
- Family III. "Blattabacteriaceae"
  - Genus I. *Blattabacterium*

Class III. "Sphingobacteria"

Order I. "Sphingobacteriales"
- Family I. *Sphingobacteriaceae*
  - Genus I. *Sphingobacterium*
  - Genus II. *Pedobacter*
- Family II. "Saprospiraceae"
  - Genus I. *Saprospira*
  - Genus II. *Haliscomenobacter*
  - Genus III. *Lewinella*
- Family III. "Flexibacteraceae"
  - Genus I. *Flexibacter*
  - Genus II. *Cyclobacterium*
  - Genus III. *Cytophaga*
  - Genus IV. *Flectobacillus*
  - Genus V. *Hymenobacter*
  - Genus VI. *Meniscus*
  - Genus VII. *Microscilla*
  - Genus VIII. *Runella*
  - Genus IX. *Spirosoma*
  - Genus X. *Sporocytophaga*
- Family IV. "Flammeovirgaceae"
  - Genus I. *Flammeovirga*
  - Genus II. *Flexithrix*
  - Genus III. *Persicobacter*
  - Genus IV. *Thermonema*
- Family V. *Crenotrichaceae*
  - Genus I. *Crenothrix*
  - Genus II. *Chitinophaga*
  - Genus III. *Rhodothermus*
  - Genus IV. *Toxothrix*

Phylum BXXI. *Fusobacteria*

Class I. "Fusobacteria"

Order I. "Fusobacteriales"
- Family I. "Fusobacteriaceae"
  - Genus I. *Fusobacterium*
  - Genus II. *Ilyobacter*
  - Genus III. *Leptotrichia*
  - Genus IV. *Propionigenium*
  - Genus V. *Sebaldella*
  - Genus VI. *Streptobacillus*
- Family II. Incertae sedis
  - Genus I. *Cetobacterium*

Phylum BXXII. *Verrucomicrobia*

Class I. *Verrucomicrobiae*

Order I. *Verrucomicrobiales*
- Family I. *Verrucomicrobiaceae*
  - Genus I. *Verrucomicrobium*
  - Genus II. *Prosthecobacter*

Phylum BXXIII. *Dictyoglomi*

Class I. "Dictyoglomi"

Order I. "Dictyoglomales"
- Family I. "Dictyoglomaceae"
  - Genus I. *Dictyoglomus*

Appendix V

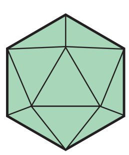
Classification of Viruses

In this appendix selected groups of viruses are briefly described and a sketch of each is included. The illustrations are not to scale but should provide a general idea of the morphology of each group. The viruses are separated into six sections based on their host preferences. The following material has been adapted with permission from chapter 21 of *Introduction to Modern Virology,* 4th ed. by N. J. Dimmock and S. B. Primrose, copyright © 1994 by Blackwell Scientific Publications, Ltd.

Viruses Multiplying in Vertebrates and Other Hosts

Note that some or all of the *Reoviridae, Bunyaviridae, Rhabdoviridae,* and *Togaviridae* multiply in both vertebrates and other hosts. Other families include genera that multiply solely in vertebrates.

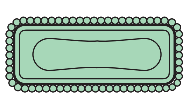
1. Family: *Iridoviridae*



Icosahedral particle (125 to 300 nm) consisting of a spherical nucleocapsid surrounded by lipid modified by protein subunits. Envelope may be present but is not required for infectivity. Double-stranded DNA of mol wt 100 to 250 × 10^6, which is circularly permuted with direct terminal repeats. Contain several enzymes. Transcription and DNA synthesis are nuclear. mRNAs have no polyA tails. Cytoplasmic.

Selected genera:
- *Iridovirus* (small 120 nm blue iridescent viruses of insects)
- *Chloriridovirus* (large 180 nm iridescent viruses of insects)
- *Ranavirus* (frog virus group)
- *Lymphocystivirus* (viruses of fish)

2. Family: *Poxviridae*



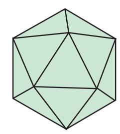
Double-stranded DNA of mol wt 85 to 240 × 10^6 with inverted terminal repeats. Largest viruses 170 to 260 × 300 to 450 nm. Complex structure composed of several layers and includes lipid. Core contains all enzymes required for mRNA synthesis. Cytoplasmic multiplication.

Selected genera:

Poxviruses of vertebrates:
- *Orthopoxvirus* (vaccinia and related viruses)
- *Molluscipoxvirus* (*Molluscum contagiosum* virus of humans)
- *Avipoxvirus* (fowlpox and related viruses)
- *Parapoxvirus* (milker's node and related viruses)

Poxviruses of insects:
- *Entomopoxvirus*

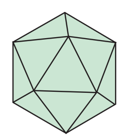
3. Family: *Parvoviridae*



Single-stranded DNA of mol wt 1.5 to 2.0 × 10^6. Particle is an 18 to 22 nm icosahedron. Contains no enzymes. Multiplication has a nuclear stage.

Genera:
- *Parvovirus*—viruses of vertebrates including humans. Virions mostly –DNA.
- *Dependovirus*—adeno-associated virus. Infects vertebrates. Particles contain either +DNA or –DNA, which forms a double strand upon extraction. Require helper adenovirus or herpesvirus for efficient replication.
- *Densovirus*—viruses of insects. Virions –DNA or +DNA. Helper not required.

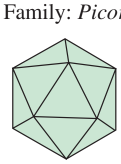
4. Family: *Reoviridae*



Ten to twelve segments of double-stranded RNA of total mol wt 12 to 20 × 10^6. Particle is a 60 to 80 nm icosahedron. Has an isometric nucleocapsid with transcriptase activity. Cytoplasmic multiplication.

Selected genera:
- *Orthoreovirus*—of vertebrates
- *Orbivirus*—of vertebrates, but also multiply in insects
- *Coltivirus*—of vertebrates and ticks
- *Rotavirus*—of vertebrates
- *Cypovirus*—cytoplasmic polyhedrosis viruses of insects
- *Phytoreovirus*—wound tumor virus

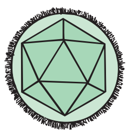
5. Family: *Picornaviridae*



Single-stranded, positive-sense RNA of mol wt 2.5 × 10^6. Particle is 30 nm and icosahedral. Multiplication is cytoplasmic.

Genera:
- *Enterovirus* (acid-resistant, primarily viruses of gastrointestinal tract)
- *Hepatovirus* (hepatitis A virus group)
- *Rhinovirus* (acid-labile, mainly viruses of upper respiratory tract)
- *Aphthovirus* (foot-and-mouth disease virus)
- *Cardiovirus* (EMC virus of mice)
- Also various viruses of insects

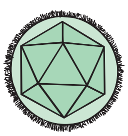
6. Family: *Togaviridae*



Single-stranded, positive-sense RNA of mol wt 4×10^6. Enveloped particles 60 to 70 nm diameter contain an icosahedral nucleocapsid. Hemagglutinate. Cytoplasmic, budding from plasma membrane. Have a subgenomic mRNA.

Genera:
Alphavirus (arboviruses, e.g., Semliki Forest and Sindbis viruses)
Rubivirus (rubella virus)

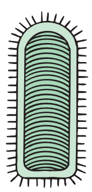
7. Family: *Flaviviridae*



Single-stranded RNA of mol wt $\sim 4 \times 10^6$. Enveloped particles 40 to 60 nm diameter. Differ from *Alphaviridae* by presence of a matrix protein, the lack of intracellular subgenomic mRNAs, and budding from the endoplasmic reticulum. Hemagglutinate. Cytoplasmic.

Genera:
Flavivirus (arboviruses, e.g., yellow fever virus)
Pestivirus (e.g., hog cholera)
Hepatitis C virus group

8. Family: *Rhabdoviridae*



Single-stranded, negative-sense RNA of mol wt 3.5 to 4.6×10^6 complementary to mRNA. The bullet-shaped or bacilliform (100 to 430 × 70 nm) particle is enveloped with 5 to 10 nm spikes. Inside is a helical nucleocapsid with transcriptase activity. Cytoplasmic, budding from plasma membrane. Some are arboviruses.

Genera:
Vesiculovirus (vesicular stomatitis virus group) viruses of vertebrates and insects
Lyssavirus (viruses of vertebrates, e.g., rabies, and of insects, e.g., sigma)
Plant viruses (e.g., lettuce necrotic yellows, potato yellow dwarf, and many others)

9. Family: *Bunyaviridae*

Three segments (large, medium, and small) of single-stranded RNA of total mol wt 6×10^6. Enveloped 100 nm particles with spikes and three internal ribonucleoprotein filaments 2 nm wide. Cytoplasmic, budding from the Golgi apparatus. Arthropod-transmitted except *Hantavirus* genus.

Selected genera:
Bunyavirus (Bunyamwera and 150 or so related viruses)
Hantavirus (Korean hemorrhagic fever or Hantaan virus) not arboviruses
Tospovirus (tomato spotted wilt group)

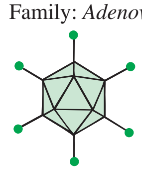
Viruses Multiplying Only in Vertebrates

1. Family: *Herpesviridae*

Double-stranded DNA of mol wt 80 to 150×10^6. Particle is a 100 to 110 nm icosahedron enclosed in a lipid envelope (120 to 200 nm virion diameter). Buds from nuclear membrane. Latency for the lifetime of the host is common.

Subfamily: *Alphaherpesvirinae*
Genera:
Simplexvirus—human (alpha) herpesvirus 1 and 2 (herpes simplex virus types 1 and 2)
Varicellovirus—human (alpha) herpesvirus 3 (varicella-zoster virus)
Subfamily: *Betaherpesvirinae* (cytomegaloviruses)
Genera:
Cytomegalovirus—human cytomegalovirus group, e.g., human (beta) herpesvirus 5 (human cytomegalovirus)
Muromegalovirus—mouse cytomegalovirus group
Subfamily: *Gammaherpesvirinae* (lymphoproliferative virus group)
Genera:
Lymphocryptovirus—e.g., human (gamma) herpesvirus 4 (Epstein-Barr virus)

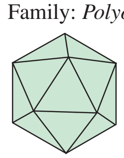
2. Family: *Adenoviridae*



Double-stranded DNA of mol wt 20 to 30×10^6. Particle is a 70 to 90 nm icosahedron, which replicates and is assembled in the nucleus.

Genera:
Mastadenovirus (adenoviruses of mammals)
Aviadenovirus (adenoviruses of birds)

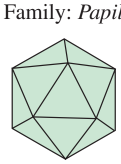
3. Family: *Polyomaviridae*



Double-stranded circular DNA. Nonenveloped, icosahedral particles are 40 nm in diameter, have 72 capsomers in a skewed arrangement, and are assembled in the nucleus. Oncogenic.

Genus:
Polyomavirus is found in rodents, humans, and other primates; DNA 3×10^6 mol wt; includes simian virus 40 (SV-40) and polyomavirus itself.

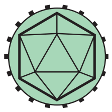
4. Family: *Papillomaviridae*



Double-stranded circular DNA. Nonenveloped, icosahedral particles are 55 nm in diameter, have 72 capsomers in a skewed arrangement, and are assembled in the nucleus. Oncogenic.

Genus:
Papillomavirus produces warts and papillomas in several mammalian species; DNA 5×10^6 mol wt; includes human papillomavirus (HPV).

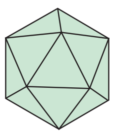
5. Family: *Hepadnaviridae*



One complete DNA minus strand of mol wt 1×10^6 with a 5′ terminal protein. DNA is circularized by an incomplete plus strand of variable length (50 to 100%), which overlaps the 3′ and 5′ termini of DNA minus. There is a 40–48 nm enveloped particle containing a core with DNA polymerase and protein kinase activities. Reverse transcriptase is involved in virus reproduction. Includes hepatitis B of humans, Pekin duck hepatitis, beechy ground squirrel hepatitis, and woodchuck hepatitis viruses. HBV is strongly associated with liver cancer.

Genus:
Orthohepadnavirus—hepatitis B virus

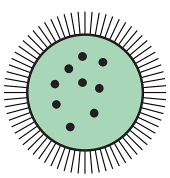
6. Family: *Caliciviridae*



Single-stranded, positive-sense RNA of mol wt 2.7×10^6. Icosahedral 37 nm particle with calixlike (cup-shaped) surface depressions.

Genus:
Calicivirus (vesicular exanthema of swine virus, Norwalk virus, possibly hepatitis E virus)

7. Family: *Arenaviridae*



Two (a large and a small) segments of single-stranded RNA of mol wt 3×10^6 and 1.3×10^6. Latter RNA is ambisense. Enveloped 50 to 300 nm particles with spikes. Contain ribosomes which have no known function. Cytoplasmic multiplication; buds from plasma membrane.

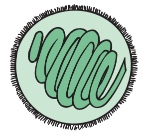
Genus:
Arenavirus (lymphocytic choriomeningitis virus and related viruses—e.g., Lassa, Junin, and Macupo viruses)

8. Family: *Paramyxoviridae*

Single-stranded RNA of mol wt 5 to 7×10^6. Enveloped 150 nm particles have spikes and contain a helical nucleocapsid 12 to 17 nm in diameter with transcriptase activity. Filamentous forms common. Cytoplasmic, budding from plasma membrane. Airborne transmission.

Subfamily: *Paramyxovirinae*
Genera:
Rubulavirus (mumps, Newcastle disease virus, parainfluenza virus) Only this genus has a neuraminidase activity which is on the same protein (HN) as the hemagglutination activity.
Morbillivirus (measles virus, canine distemper) Hemagglutinate
Subfamily: *Pneumovirinae*
Pneumovirus (respiratory syncytial virus group)

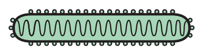
9. Family: *Orthomyxoviridae*



Eight segments of single-stranded RNA of total mol wt 4×10^6. Enveloped 100 nm particles have spikes and contain a helical nucleocapsid 9 nm in diameter with transcriptase activity. Only A and B virions have separate hemagglutinin and neuraminidase proteins. Multiplication requires the nucleus. RNA segments in a mixed infection readily assort to form genetically stable hybrids within a virus. Buds from plasma membrane.

Genera:
Influenza virus A and B
Influenza C virus. Has seven RNA segments and a receptor-destroying activity (a sialic acid-O-acetyl esterase), which is on the hemagglutinin protein.

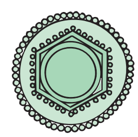
10. Family: *Filoviridae*



Long filamentous particles 800 to 900 (sometimes 14,000) × 80 nm with helical nucleocapsid of 50 nm diameter. Linear, single-stranded, negative-sense RNA of mol wt 4.2×10^6. Buds from plasma membrane. Contains Marburg and Ebola viruses, which are highly pathogenic for humans. Transmitted by contact.

Genus:
Filovirus—Marburg and Ebola viruses

11. Family: *Retroviridae*



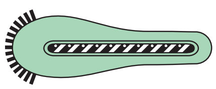
Single-stranded "diploid" RNA with unique sequence having a mol wt 1 to 3×10^6. Enveloped 80–100 nm particles with a core containing a helical nucleoprotein. Contains RNA-dependent DNA polymerase (reverse transcriptase). The DNA provirus is nuclear.

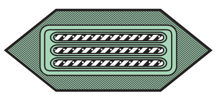
Selected genera:
MLV-related viruses (mammalian type C retrovirus group)—murine leukemia virus
Spumavirus—human foamy virus
HTLV-BLV group—human T-cell lymphotrophic viruses types 1 and 2
Lentivirus—human immunodeficiency viruses types 1 and 2; simian immunodeficiency virus, visna virus

Viruses Multiplying Only in Invertebrates

Viruses occur not only in insects, crustacea, and mollusks but probably in all groups of invertebrates. The *Poxviridae, Reoviridae, Parvoviridae, Rhabdoviridae,* and *Togaviridae* (see earlier) have representatives that multiply in invertebrates. Some plant viruses are transmitted by, but do not multiply in, these vectors.

1. Family: *Baculoviridae*

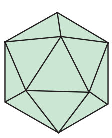




Double-stranded circular DNA of mol wt 60 to 110 × 10^6. Bacilliform particles 30 to 60 nm × 250 to 300 nm with an outer membrane. May be occluded in a protein inclusion body containing usually one particle (granulosis viruses, upper illustration) or in a polyhedra containing many particles (polyhedrosis viruses, lower illustration).

Selected genera:
 Nuclear polyhedrosis virus
 Granulosis viruses
 Nonoccluded baculoviruses

2. Family: *Tetraviridae*

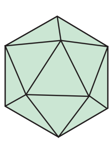


Single-stranded RNA of mol wt 1.8 × 10^6 in a 35 nm particle. T = 4 (whereas in *Picornaviridae* T = 1). All isolated from Lepidoptera. No infection of cultured cells. *Nudaurelia* β virus.

Viruses Multiplying Only in Plants

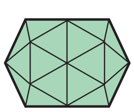
Knowledge of virus multiplication in plants is relatively rudimentary since the cell culture systems are less manageable than animal cell cultures. Work has concentrated on physical properties and disease characteristics. The *Reoviridae* and *Rhabdoviridae* have members that multiply in both plants and invertebrates. The plant viruses listed on the following page are not known to multiply in their invertebrate vector. Differences in virus proteins and translation strategy are important classification criteria.

1. *Caulimoviridae*



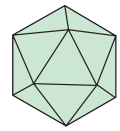
Open circular dsDNA with single-strand discontinuities like the hepadnavirus. Mol wt 4 to 5 × 10^6. Isometric 50 nm particles; some are bacilliform. Contains reverse transcriptase. Aphid vectors. Includes cauliflower mosaic virus.

2. *Geminiviridae*



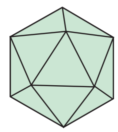
Circular single-stranded DNA of mol wt 0.7 to 0.8 × 10^6. Two incomplete icosahedral particles joined as a pair, ~18 × 30 nm, and usually found in the nucleus. One or two molecules of DNA per pair of particles. Persistent in whitefly or leafhopper vectors.

3. *Luteoviridae*



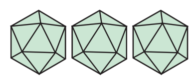
Single-stranded, positive-sense RNA of mol wt 2 × 10^6. Isometric 25 to 30 nm particle. Persistent retention by aphid vectors. Includes barley yellow dwarf virus.

4. *Tombusviridae*



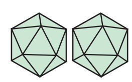
Single-stranded, linear, positive-sense RNA of mol wt 1.5 × 10^6. Not polyadenylated. Particle 32 to 35 nm in diameter. Cytoplasmic, nuclear, and sometimes mitochondrial location. Transmitted through the soil and by mechanical inoculation, contact, and seeds. Includes tobacco necrosis virus and tomato bushy stunt virus.

5. *Bromoviridae*



Three single-stranded, linear, positive-sense RNAs. Icosahedral particles 26 to 35 nm diameter and bacilliform particles (18 to 26 nm × 30 to 85 nm). Infectivity requires the three RNAs. Each is in a different particle. Assembly in cytoplasm. Some with a beetle vector. Includes brome mosaic virus, cucumber mosaic virus, and alfalfa mosaic virus.

6. *Comoviridae*

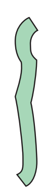


Two 28 to 30 nm nonenveloped particles containing single-stranded, linear, positive-sense RNA of mol wt 2.4 or 1.4 × 10^6. Two coat polypeptides encoded by the smaller RNA. Both RNAs needed for infectivity. Cytoplasmic. Transmitted by beetles or seed. Includes cowpea mosaic virus and tobacco ringspot virus.

7. *Tobamovirus*—tobacco mosaic virus group

Single-stranded, linear, positive-sense RNA of mol wt 2 × 10^6. Rigid cylindrical particles 300 × 18 nm. Transmitted mechanically or by seed.

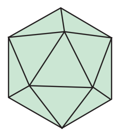
8. *Potyviridae*



Single-stranded, linear, positive-sense RNA of mol wt 3.0 to 3.5 × 10^6. Particle is a flexuous rod with helical symmetry, usually 650 to 900 × 11 to 15 nm. Cytoplasmic but some have nuclear inclusions. Nonpersistent in aphid vectors. Includes potato virus y.

Viruses Multiplying Only in Algae, Fungi, and Protozoa

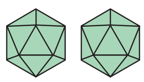
1. Family: *Totiviridae*



Isometric 40 to 43 nm particles with a genome consisting of a single molecule of double-stranded RNA of mol wt 3.3 to 4.2×10^6. One major capsid protein. Single shell. Transcriptase present. Cytoplasmic.

Genus:
Totivirus (*Saccharomyces cerevisiae* virus LA)

2. Family: *Phycodnaviridae*

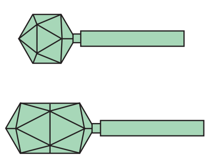


Large polyhedral 130 to 200 nm diameter particles containing linear, double-stranded DNA of mol wt 150 to 210 $\times 10^6$. Nonenveloped. Contain lipid in shell. Infect *Paramecium* and *Chlorella* sp.

Viruses Multiplying Only in Bacteria

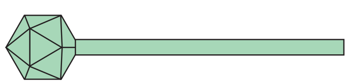
Surprisingly little is known of the comparative biology of bacterial viruses as only a few representatives have been studied in detail.

1. Family: *Myoviridae* (Phages with contractile tails)



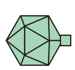
Linear double-stranded DNA of mol wt 120×10^6. Head isometric or elongated, 110×80 nm; complex contractile tail 113 $\times$ 16 nm. Tail has collar, base plate, spikes, and fibers. Includes T2, T4, T6, PBS1, SP8, SP50, P2, Mu.

2. Family: *Siphoviridae* (Phages with long, noncontractile tails)



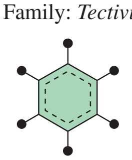
Linear double-stranded DNA of mol wt 33×10^6. Head 60 nm diameter; long noncontractile tail up to 570 nm. No breakdown of host DNA. Includes λ, χ (chi), and ϕ80.

3. Family: *Podoviridae* (Phages with short tails)



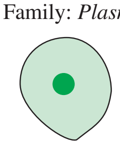
Linear double-stranded DNA of mol wt 25×10^6. Head 60 nm diameter. Short (17×8 nm) tail with 6 short fibers. Includes T7 and P22.

4. Family: *Tectiviridae* (Phages with double capsids)



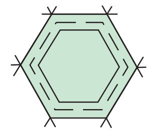
Linear double-stranded DNA of mol wt 10×10^6 in 63 nm particles. Contains internal lipid. Double capsid with a rigid outer shell and flexible inner coat. After injection of DNA a tail structure of about 60 nm appears. PRD1, Bam35.

5. Family: *Plasmaviridae* (Pleomorphic phages)



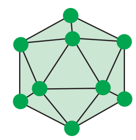
Circular double-stranded DNA of 8×10^6 mol wt in lipid-containing envelope. 50 to 125 nm particle with small dense core. Slightly pleomorphic. Formed by budding. Infects mycoplasmas.

6. Family: *Corticoviridae* (PM2 phage group)



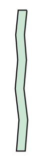
Circular double-stranded DNA of mol wt 6×10^6, isometric 60 nm particle, lipid between protein shells, no envelope, no tail. Spikes at vertices. Infects *Pseudomonas.*

7. Family: *Microviridae* (Isometric phages with ssDNA)



Circular single-stranded DNA of mol wt 1.6 to 1.7×10^6. Icosahedron (25 to 27 nm) with knobs on 12 vertices. No envelope. Includes ϕX174 and G4.

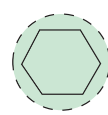
8. Family: *Inoviridae* (Rod-shaped phages)



Helical, filamentous or rod-shaped phages with a circular, single-stranded, positive-sense DNA. No host lysis.

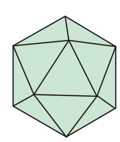
Genera:
Inovirus
DNA of 1.9 to 2.7×10^6, long flexible filamentous particle 760 to $1{,}950 \times 6$ to 8 nm. Host bacteria not lysed. Includes M13 and fd phages.
Plectovirus
Mycoplasma virus type 1 phages. DNA 2.5 to 5.2×10^6 mol wt. Short rods of 85 to 280×14 nm.

9. Family: *Cystoviridae* (ϕ6 phage group)



Three molecules of linear, double-stranded RNA of mol wt 2.3, 3.1, and 5.0×10^6. Isometric 75 nm particle with lipid envelope. Infects *Pseudomonas.*

10. Family: *Leviviridae* (single-stranded RNA phages)



Linear, single-stranded, positive-sense RNA of mol wt 1.2×10^6. Icosahedral capsid is 24 nm.

Genera:
Levivirus—coliphage MS2 group
Allolevivirus—coliphage Qβ group

CREDITS

Photos

Chapter 1

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Chapter 2

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Illustrators

Precision Graphics

11.9b, 12.20, 12.24, 15.2b,c, 19.8, 19.12d, p. 451, p. 487, p. 518, 22.1, 22.11, 22.16b, 22.30, p. 537, 23.1b, p. 537, 24.3a, p. 553, p. 571, p. 584, 29.7b, 29.8a, 29.15, 29.25b, 29.27, 30.3, 30.11, 31.17, 42.13, 42.21

Nobel Prizes Awarded for Research in Microbiology

Date	Scientist[a]	Research
1901	E. von Behring (GR)	Diphtheria antitoxin
1902	R. Ross (GB)	Cause and transmission of malaria
1905	R. Koch (GR)	Tuberculosis research
1907	C. Laveran (F)	Role of protozoa in disease
1908	P. Ehrlich (GR) E. Metchnikoff (R)	Work on immunity
1913	C. Richet (F)	Work on anaphylaxis
1919	J. Bordet (B)	Discoveries about immunity
1928	C. Nicolle (F)	Work on typhus fever
1930	K. Landsteiner (US)	Discovery of human blood groups
1939	G. Domagk (GR)	Antibacterial effect of prontosil
1945	A. Fleming (GB) E. B. Chain (GB) H. W. Florey (AU)	Discovery of penicillin and its therapeutic value
1951	M. Theiler (SA)	Development of yellow fever vaccine
1952	S. A. Waksman (US)	Discovery of streptomycin
1954	J. F. Enders (US) T. H. Weller (US) F. Robbins (US)	Cultivation of poliovirus in tissue culture
1957	D. Bovet (I)	Discovery of the first antihistamine
1958	G. W. Beadle (US) E. L. Tatum (US) J. Lederberg (US)	Microbial genetics
1959	S. Ochoa (US) A. Kornberg (US)	Discovery of enzymes catalyzing nucleic acid synthesis
1960	F. M. Burnet (AU) P. B. Medawar (GB)	Discovery of acquired immune tolerance to tissue transplants
1962	F. H. C. Crick (GB) J. D. Watson (US) M. Wilkins (GB)	Discoveries concerning the structure of DNA
1965	F. Jacob (F) A. Lwoff (F) J. Monod (F)	Discoveries about the regulation of genes
1966	F. P. Rous (US)	Discovery of cancer viruses
1968	R. W. Holley (US) H. G. Khorana (US) M. W. Nirenberg (US)	Deciphering of the genetic code
1969	M. Delbrück (US) A. D. Hershey (US) S. E. Luria (US)	Discoveries concerning viruses and viral infection of cells
1972	G. Edelman (US) R. Porter (GB)	Research on the structure of antibodies
1975	H. Temin (US) D. Baltimore (US) R. Dulbecco (US)	Discovery of RNA-dependent DNA synthesis by RNA tumor viruses; reproduction of DNA tumor viruses
1976	B. Blumberg (US) D. C. Gajdusek (US)	Mechanism for the origin and dissemination of hepatitis B virus; research on slow virus infections
1977	R. Yalow (US)	Development of the radioimmunoassay technique
1978	H. O. Smith (US) D. Nathans (US) W. Arber (SW)	Discovery of restriction enzymes and their application to the problems of molecular genetics
1980	B. Benacerraf (US) G. Snell (US) J. Dausset (F)	Discovery of the histocompatibility antigens
	P. Berg (US) W. Gilbert (US) & F. Sanger (GB)	Development of recombinant DNA technology (Berg); development of DNA sequencing techniques (Chemistry Prize)
1982	A. Klug (GB)	Development of crystallographic electron microscopy and the elucidation of the structure of viruses and other nucleic-acid–protein complexes (Chemistry Prize)
1984	C. Milstein (GB) G. J. F. Kohler (GR) N. K. Jerne (D)	Development of the technique for formation of monoclonal antibodies (Milstein & Kohler); theoretical work in immunology (Jerne)
1986	E. Ruska (GR)	Development of the transmission electron microscope (Physics Prize)
1987	S. Tonegawa (J)	The genetic principle for generation of antibody diversity
1988	J. Deisenhofer, R. Huber, and H. Michel (GR)	Crystallization and study of the photosynthetic reaction center from a bacterial membrane
	G. Elion (US) G. Hitchings (US)	Development of drugs for the treatment of cancer, malaria, and viral infections
1989	J. M. Bishop (US) H. E. Varmus (US)	Discovery of oncogenes
	S. Altman (US) T. R. Cech (US)	Discovery of catalytic RNA
1993	K. B. Mullis (US)	Invention of the polymerase chain reaction
	M. Smith (US)	Development of site-directed mutagenesis
	R. J. Roberts (US) P. A. Sharp (US)	Discovery of split genes
1996	P. C. Doherty (AU) R. M. Zinkernagel (SW)	Discovery of the mechanism by which T lymphocytes recognize virus-infected cells
1997	S. Prusiner (US)	Discovery of prions

[a]The Nobel laureates were citizens of the following countries: Australia (AU), Belgium (B), Denmark (D), France (F), Germany (GR), Great Britain (GB), Italy (I), Japan (J), Russia (R), South Africa (SA), Switzerland (SW), and the United States (US).

Comparison of *Bacteria, Archaea,* and *Eucarya*

Property	*Bacteria*	*Archaea*	*Eucarya*
Membrane-Enclosed Nucleus with Nucleolus	Absent	Absent	Present
Complex Internal Membranous Organelles	Absent	Absent	Present
Cell Wall	Almost always have peptidoglycan containing muramic acid	Variety of types, no muramic acid	No muramic acid
Membrane Lipid	Have ester-linked, straight-chained fatty acids	Have ether-linked, branched aliphatic chains	Have ester-linked, straight-chained fatty acids
Gas Vesicles	Present	Present	Absent
Transfer RNA	Thymine present in most tRNAs	No thymine in T or TψC arm of tRNA	Thymine present
	N-formylmethionine carried by initiator tRNA	Methionine carried by initiator tRNA	Methionine carried by initiator tRNA
Polycistronic mRNA	Present	Present	Absent
mRNA Introns	Absent	Absent	Present
mRNA Splicing, Capping, and Poly A Tailing	Absent	Absent	Present
Ribosomes			
Size	70S	70S	80S (cytoplasmic ribosomes)
Elongation factor 2	Does not react with diphtheria toxin	Reacts	Reacts
Sensitivity to chloramphenicol and kanamycin	Sensitive	Insensitive	Insensitive
Sensitivity to anisomycin	Insensitive	Sensitive	Sensitive
DNA-Dependent RNA Polymerase			
Number of enzymes	One	Several	Three
Structure	Simple subunit pattern (4 subunits)	Complex subunit pattern similar to eucaryotic enzymes (8–12 subunits)	Complex subunit pattern (12–14 subunits)
Rifampicin sensitivity	Sensitive	Insensitive	Insensitive
Polymerase II Type Promoters	Absent	Present	Present
Metabolism			
Similar ATPase	No	Yes	Yes
Methanogenesis	Absent	Present	Absent
Nitrogen fixation	Present	Present	Absent
Chlorophyll-based photosynthesis	Present	Absent	Present[a]
Chemolithotrophy	Present	Present	Absent

[a]Present in chloroplasts (of bacterial origin).

(repeated as Table 19.8)